



الاتحاد للمدفوعات
AI Etihad Payments

**Jaywan eCom DS Specification Document and
Implementation Guidelines 1.0**



Version

Version	Release Date	Comments
1.0	12-Aug-24	Initial Release

Disclaimer

This document, including any referenced documents in the appendices and any amended versions shall be at all times the property of the EI Etihad Payments (AEP), who shall also own the copyright of these documents. All brand or product names mentioned are trademarks or registered trademarks of their respective holders. These documents may not be reproduced, transmitted electronically or otherwise, in whole or in part, without the prior written permission of AEP.

AI Etihad Payments is the national payments entity created in 2023 as part of the National Payment Systems Strategy to develop and operate best-in-class financial markets infrastructure for the UAE. It is a fully owned subsidiary of The Central Bank of UAE. AEP is formed to Establish and operationalize best-in-class payment services and infrastructure, contributing to financial stability and financial inclusion in the country.

The UAE Domestic Card scheme (Jaywan) is one of the initiatives of the CBUAE National Payment System Strategy (NPSS). This initiative, launched in 2024, aims to enhance the country's financial infrastructure by providing a secure, efficient, and cost-effective card payment method for both consumers and merchants within the UAE. The primary goals of the Jaywan scheme include reducing reliance on international payment networks, keeping transaction fees within the country, and enhancing financial inclusion making

payment services more accessible to all segments of the population by keeping the transaction fees low when compared to international schemes.

Table of Contents

Version	2
Disclaimer	2
1 Introduction	6
2 Glossary	7
3 Convention	10
4 References	10
5 Objective of the document	10
6 Confidentiality and Copyright Notice	11
7 JaywanDS	11
7.1 Context.....	11
7.1.1 About JaywanDS	12
7.1.2 JaywanDS features	12
7.2 JaywanDS and EMV 3DS	14
7.2.1 Customer Journey	16
7.2.2 Participants	17
7.2.3 Components	19
7.2.4 3DS message types	21
7.2.5 Authentication flows.....	24
7.2.6 Authentication channels	28
7.2.7 Transaction Status Indicators.....	29
7.2.8 Checkout UI template	31
7.3 CAVV computation.....	35
7.3.1 CAVV generation.....	35
7.3.2 CAVV calculation	36
7.3.3 Authentication Result Code Derivation	37
8 Transaction Flow	39
8.1 Payment Transaction.....	41
8.1.1 JaywanDS - Challenge Authentication Purchase Transaction.	41
8.1.2 JaywanDS - Challenge Authentication Purchase Transaction (3DS SDK Flow)	43

8.1.3	JaywanDS - FrictionLess Purchase Transaction.	45
8.2	3DS method.....	46
9	Tokenization	50
9.1	Tokenized Payment Flow.	51
9.1.1	JaywanDS - Tokenized Challenge Flow.	52
9.1.2	JaywanDS - Tokenized Frictionless Flow.	54
10	Project Implementation for Participants	56
10.1	Prerequisites.....	56
10.2	JaywanDS OnBoarding Process	57
10.2.1	Registration	57
10.2.2	Certification Process.....	58
10.2.3	Deployment	59
10.3	ACS Onboarding information	60
10.3.1	ACS Reference Number	60
10.3.2	Card range data	60
10.3.3	ACS Operator ID	61
10.3.4	Signed Certificate for TLS connection.....	61
10.3.5	ACS signed content (Certificate for JWS).....	67
10.4	3DSS Onboarding information	70
10.4.1	3DS Server Reference Number	70
10.4.2	Card Range Data.....	71
10.4.3	3DS Server Operator ID	71
10.4.4	3DS Requestor Name.....	71
10.4.5	Acquirer BIN.....	72
10.4.6	Signed Certificate for TLS connexion.....	72
11	Timeouts	76
11.1	Transaction Timeout	76
11.2	Connection Timeout	76
11.3	Read Timeout.....	76
11.4	Timeout Values.....	77
12	Error Codes	78
12.1	Error Component.....	78
12.2	Error Codes	78
13	Sample Messages (3DSS & ACS)	79
13.1	Message for 3DSS.....	79

13.1.1	PREQ Message from 3DSS - DS	79
13.1.2	PRES Message from DS - 3DSS.....	80
13.1.3	AREQ Message from 3DSS - DS (Browser Flow)	80
13.1.4	ARES Message from DS - 3DSS (Browser Flow).....	82
13.1.5	RREQ Message from DS - 3DSS (Browser Flow)	83
13.1.6	RRES Message from 3DSS - DS (Browser Flow).....	83
13.1.7	AREQ Message from 3DSS - DS (APP Flow)	84
13.1.8	ARES Message from DS -3DSS (APP Flow)	88
13.1.9	RREQ Message from DS - 3DSS (App Flow)	90
13.1.10	RRES Message from 3DSS - DS (App Flow)	91
13.2	Message for ACS	91
13.2.1	AREQ Message from DS - ACS (APP Flow)	91
13.2.2	ARES Message from ACS - DS (APP Flow).....	95
13.2.3	RREQ Message from ACS - DS (APP Flow)	96
13.2.4	RRES Message from DS -ACS (App Flow).....	97
13.2.5	AREQ Message from DS - ACS (Browser Flow)	97
13.2.6	ARES Message from ACS - DS (Browser Flow).....	99
13.2.7	RREQ Message from ACS - DS (Browser Flow)	100
13.2.8	RRES Message from DS - ACS (Browser Flow).....	100
14	Annexure A - EMVCo Reference Guide	101
15	Annexure B - Token Message Extension.....	101

1 Introduction

AI Etihad Payments is the national payments entity created in 2023 as part of the National Payment Systems Strategy to develop and operate best-in-class financial markets infrastructure for the UAE. It is a fully owned subsidiary of The Central Bank of UAE. AEP is formed to Establish and operationalize best-in-class payment services and infrastructure, contributing to financial stability and financial inclusion in the country.

The UAE Domestic Card scheme (Jaywan) is one of the initiatives of the CBUAE National Payment System Strategy (NPSS). This initiative, launched in 2024, aims to enhance the country's financial infrastructure by providing a secure, efficient, and cost-effective card payment method for both consumers and merchants within the UAE. The primary goals of the Jaywan scheme include reducing reliance on international payment networks, keeping transaction fees within the country, and enhancing financial inclusion making payment services more accessible to all segments of the population by keeping the transaction fees low when compared to international schemes.

This document, including any referenced documents in the appendices and any amended versions shall always be the property of the EI Etihad Payments (AEP), who shall also own the copyright of these documents. All brand or product names mentioned are trademarks or registered trademarks of their respective holders. These documents may not be reproduced, transmitted electronically or otherwise, in whole or in part, without the prior written permission of AEP.

2 Glossary

Terminology	Abbreviation	Definition
3DS requestor environment	-	The 3DS Requestor Environment is a collective term for components under the 3DS Requestor's control that support 3-D Secure. The 3DS Requestor Environment components include: ▪ 3DS Requestor (merchant/PSP) ▪ 3DS Client (Browser JS, 3DS SDK) ▪ 3DS Server
3DS Server	3DSS	Refers to the 3DS Integrator's server or systems that handle online transactions and facilitates communication between the 3DS Requestor and the DS.
Access Control Server	ACS	A component that operates in the Issuer Domain, that verifies whether authentication is available for a card number and device type and authenticates specific Cardholders.
Additional Factor of Authentication	AFA	This is getting used to authenticate the user or cardholder as a security measure, the AFA can be in any form such as some commonly used forms are PIN, dynamic one-time password (OTP), static code, etc.
Switch	SW	Switching service supports routing of interbank card transactions through Card network.
Cardholder Authentication Verification Value	CAVV	Authentication value.

Terminology	Abbreviation	Definition
Card Not Present	CNP	Card Not Present transaction, are remote transaction which does not require the card to be physically presented at the point of transactions.
Directory Server	DS	A server component operated in the Interoperability Domain; it performs a number of functions that include authenticating the 3DS Server, routing messages between the 3DS Server and the ACS, and validating the 3DS Server, the 3DS SDK, and the 3DS Requestor.
EMV 3DS SDK	SDK	3-D Secure Software Development Kit (SDK). A component that is incorporated into the 3DS Requestor App. The 3DS SDK performs functions related to 3-D Secure on behalf of the 3DS Server.
EMVCo	-	EMVCo is a global technical body that facilitates worldwide interoperability and acceptance of secure payment transactions by managing and evolving the EMV Specifications and related testing processes.
JSON Web Encryption	JWE	JSON Web Encryption (JWE) represents encrypted content using JSON-based data structures. Reference: rfc7516
Merchant APP	APP	An App on a Consumer Device that can process a 3-D Secure transaction through the use of a 3DS SDK. The merchant App is enabled through integration with the 3DS SDK.
AI Etihad Payments	AEP	AI Etihad Payments (AEP), a subsidiary of the Central Bank of the UAE (CBUAE) created in 2023 as part of National Payment Systems Strategy to develop and

Terminology	Abbreviation	Definition
		operate best-in-class financial markets infrastructure for the UAE.
Token Service Provider	TSP	Entity responsible for the issuance and management of payment tokens associated with real PAN. It tokenizes/detokenizes, manages lifecycle of token and performs cryptographic validations based on Jaywan specification.
Registered Application Provider Identifier	RID	Registered Application Provider Identifier (RID) is unique to a Payment System. RIDs are defined by the ISO 7816-5 standard and are issued by the ISO/IEC 7816-5 registration authority. RIDs are 5 bytes.
Risk Based Analysis	RBA	Risk Analysis is a proven way of identifying and assessing factors that could negatively affect the success of a transaction. It allows the issuer bank to examine the risks of an authentication/ authorization to decide whether or not to approve or decline.
Software Development Kit	SDK	A software development kit (SDK) is a set of software libraries designed to include pre-built functionalities into developer's code.

3 Convention

The key words **"MUST"**, **"MUST NOT"**, **"REQUIRED"**, **"SHALL"**, **"SHALL NOT"**, **"SHOULD"**, **"SHOULD NOT"**, **"RECOMMENDED"**, **"MAY"**, and **"OPTIONAL"** in this document are to be interpreted as described in [RFC-2119].

Annotation Convention for Attributes

Abbreviation	Meaning
N	Numeric value
AN	Alphanumeric
ANS	Alpha numeric with special characters (~!@#\$%^&*()_+ = - , . / < > ? [] \ : ; " ' Space)

4 References

Reference	Description
[EMV-3DS-CORE-SPECIFICATION]	EMV® 3-D Secure Protocol and Core Functions Specification – version 2.2.0
[EMV-3DS-EXT-BDG]	EMV® 3-D Secure - Bridging Message Extension - Version 1.0 August 2022
[EMV-3DS-EXT-TOKEN]	EMV® 3-D Secure – Payment Token Message Extension - Version 1.0 June 2021

5 Objective of the document

This document provides guidance and recommendation to the participant who want to go for the implementation and onboarding with Jaywan Directory Server(DS).

This guide provides to the participants a description of the JaywanDS ecosystem and service, the prerequisites to achieved, the list of requirements defines by JaywanDS, and the steps of the JaywanDS onboarding process.

6 Confidentiality and Copyright Notice

This document is of restricted use. No part of this document may be reproduced in any form by any means without prior written authorization from AI-Etihad Payments.

7 JaywanDS

7.1 Context

In order to provide an interoperable and a secure solution for Card Not Present (CNP) transactions, EMV 3-D Secure Protocol (EMV 3DS). It covers traditional browser and in-app flows. The EMV 3DS standard provides mechanisms to cover user identification and verification for card not present scenario, including payment and non-payment use cases.

Ecommerce transaction is composed by two different parts aiming to provide a consistent payment flow, the authentication, and the authorization.

The authentication part, or leg, is the mechanism deployed to verify the user identity and to ensure that the user is the one originating the payment with a given payment instrument. With EMV 3DS protocol, the authentication is going through the card network 3DS module called a Directory Server (DS).

The authorization part, or leg, is the mechanism whereby an issuer approves or declines a transaction initiated by a merchant and submitted by its acquirer bank. The issuer performs a risk analysis and an account balance check for funds verification to evaluate the transaction.

7.1.1 About JaywanDS

Jaywan program to implement EMV 3DS standard is named JaywanDS®, where authentication solution for ecommerce to EMV 3DS as part of evolving regulations aimed at enhancing the security of online transactions. It provides a strong security layer for user authentication and complies with EMV and regulator’s regulations

JaywanDS supports following EMV 3DS version

EMV 3DS version	Supported
2.1.0	YES
2.2.0	YES
2.3.1.1	No

7.1.2 JaywanDS features

This section aims to list the EMV 3DS features supported by JaywanDS.

Supported features	Description	Inclusion
Authentication flow		
Frictionless	Seamless authentication performed without user interaction. Authentication is based on risk assessment using data carried on EMV 3DS protocol.	✓
Challenge	Issuer interacts with the cardholder to perform the authentication. The challenge method is selected by the Issuer	✓

Supported features	Description	Inclusion
	and the challenge is initiated from the 3DS Client to the Issuer ACS.	
Challenge methods		
Static Passcode	Static secret known by the cardholder. If available, only used for non-rechargeable prepaid card or corporate prepaid card.	✓
Knowledge based Authentication (KBA)	Questions are submitted by the ACS to the cardholder. The answers to these questions have been stored during enrolment or account creation.	✓
One-Time-Password (OTP)	Dynamic value generated by the issuer and delivered to the cardholder through known channel. Following types are included: ▪ SMS OTP ▪ OTP authenticator application ▪ OTP other method (e.g. mail, key fob) IMPORTANT NOTE: SMS OTP alone shall not be considered as a secured challenge method.	✓
Out-Of-Band (OOB)	Issuer decides to use an authentication method where the value is not transmitted through EMV 3DS CReq/CRes messages. However, the authentication takes place during the transaction momentum. Following types are included ▪ Biometric method ▪ Issuer login account ▪ Issuer authenticator application	✓

Supported features	Description	Inclusion
channels		
Browser	The checkout occurs in the merchant web application through the cardholder's browser. The merchant web page includes an EMV 3DS Client (embedded JavaScript) allowing interactions with the 3DS Requestor and the ACS. Device information is gathered from browser's agents.	✓
Application	The checkout occurs in a merchant application (in-app). The merchant application integrates a 3DS SDK allowing interactions with the 3DS Requestor and the gathering of device information.	✓
3RI	The transaction is initiated without the presence of the cardholder. Also named Merchant Initiated Transaction (MIT) in opposition to Cardholder Initiated Transaction (CIT).	✓ of
Additional Issuer data collection		
3DS method	The merchant web page allows the Issuer ACS to gather device information using cardholder's browser agent. This information is used by the Issuer to improve risk assessment.	✓

7.2 JaywanDS and EMV 3DS

EMV 3-Domain Secure (EMV 3DS) is a messaging protocol for the authentication of the cardholder in a card-not-present (CNP) ecommerce transaction. It aims to improve the security, user experience and interoperability to enable safe ecommerce throughout multiple channels.

The three domains involved in the protocol are the Issuer domain, the Interoperability domain (Payment network), and the Acquirer domain.

- **The Issuer domain:** This domain is responsible for card management, including card enrolment, cardholder authentication during the authentication process, and authorization of online transactions.
- **The Interoperability domain:** This domain serves as the intermediary between the acquirer and issuer, facilitating the seamless exchange of requests and responses between their respective domains.
- **The Acquirer domain:** This domain handles merchant onboarding, initiates authentication requests during online payments made by cardholders, and receives payments for online transactions. By requesting a 3DS operation, the acquirer can shift liability for online Card-Not-Present (CNP) transactions to the issuer.

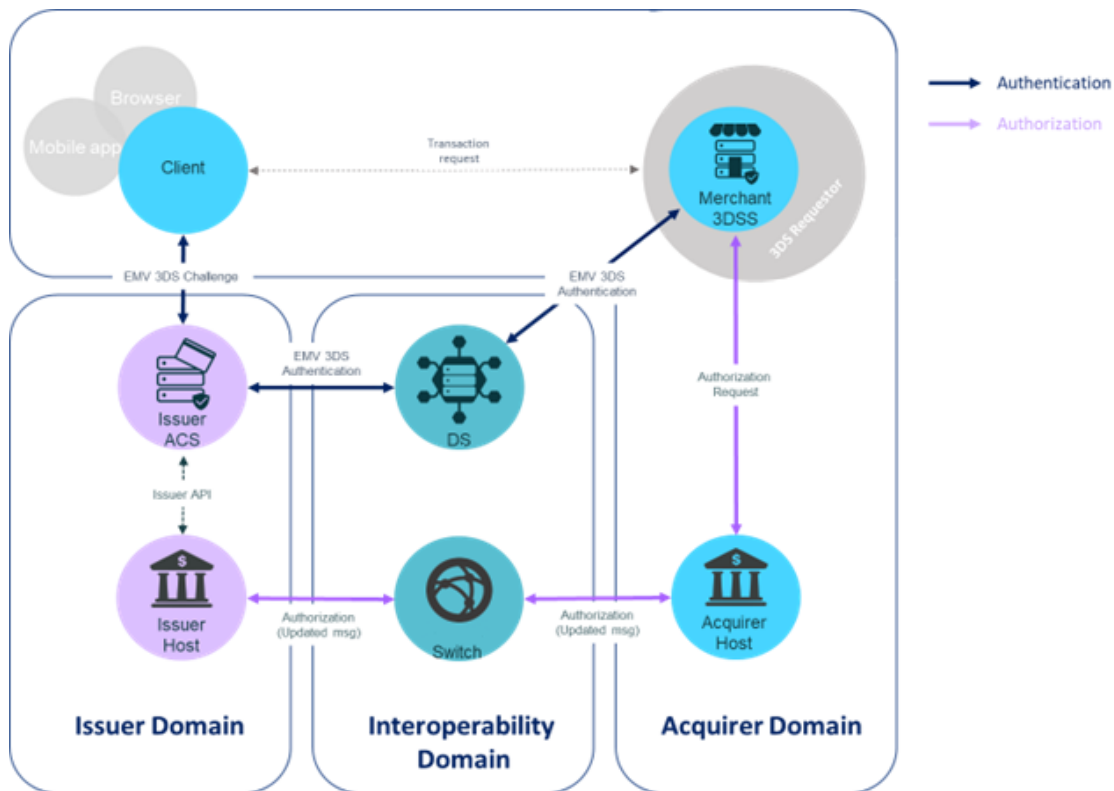
JaywanDS system, based on EMV 3DS, manages the messaging between the stakeholders processing an ecommerce transaction.

The ecosystem is designed to:

- Support a seamless user experience,
- Support a wide range of authentication methods,
- Decrease fraud levels for online transactions,
- Improve risk-based assessment for online transactions,

7.2.1 Customer Journey

The JaywanDS three domains architecture is represented in the diagram below.



Following is the customer journey to process the 3DS transactions in JaywanDS.

1. Customer enters on merchant website and initiates a purchase request at merchant checkout page
2. Customer enters their card details and initiate the payment request.
3. Merchant initiates the authentication request from the Merchant 3DSS to JaywanDS
4. JaywanDS forwards the request to issuer ACS for authentication, issuer ACS validates the user and share the response back to JaywanDS DS and further to Merchants 3DSS server.
In the same response ACS request for authentication from customer.

5. Merchant basis the authentication response, merchant route the customer to issuer URL and customer fulfils the requested information (entering OTP, passcode, etc.).
6. Post the authentication, issuer ACS sends the result of authentication to merchants 3DSS and merchant send authentication response to the Issuer ACS.
7. Post the successful authentication response, merchant from acquiring host initiates the authorisation request to issuer host for the debit to customer accounts.
8. Post debit confirmation, issuer host communicates the result to acquirer host which is further communicated to merchants.
9. Merchant displays the status of transaction and delivers the services

7.2.2 Participants

Cardholders, Issuers, Merchants, Acquirers, and a Payment Network are involved in a transaction. The table below provides an overview of the roles and responsibilities for each participant within the JaywanDS ecosystem.

Participant	Role	Responsibility
Cardholder	The Cardholder is the individual owning the payment card authorized on the Payment Network and enabled for ecommerce transactions.	The Cardholder is responsible for initiating the ecommerce transaction in the Merchant's environment.
Issuer	Represents the banking institution that manages and issues the cardholder's payment card. It is the entity in charge of verifying card, cardholder and device information to confirm or	The Issuer is responsible for: 1. The enrolment and the management of the cards that are eligible to participate in JaywanDS, 2. Configure and manage the risk-based authentication rules.

Participant	Role	Responsibility
	not the authentication through JaywanDS.	3. Validating the authorization response.
Merchant	The Merchant provides the ecommerce platform with the online shopping environment and checkout mechanism (website or application), where the cardholder can purchase goods and services.	The Merchant is responsible for: 1. Capturing the card information and providing transaction information to the Acquirer, 2. Sending the authentication request.
Acquirer	The Acquirer is the banking institution that manages the merchant's payment requests. It is in charge of sending the authorization message to the relevant Payment Networks and starting the authentication process. The Acquirer also determines the Merchant's eligibility for participating in JaywanDS program.	The Acquirer is responsible for: 1. The eligibility assessment, the enrolment, and the management of the Merchants registered in JaywanDS, 2. Authorization and clearing requests to the Issuer through the Payment Network, 3. Starting the Cardholder authentication process.
Payment Network	The Payment Network orchestrates the authentication interactions with the routing of the messages and sets the requirements for participating in JaywanDS program.	The Payment Network is responsible for: 1. Establishing and operating the authentication and authorization channel / process between the Acquirer and Issuer, 2. Defining JaywanDS rules and ecosystem, 3. Verify Issuers' and Acquirers' compliancy with applicable requirements,

Participant	Role	Responsibility
		4. Routing the authentication and authorization messages between Issuers and Acquirers.

7.2.3 Components

Issuer domain

Component	Description
Access Control Server (ACS)	The Issuer ACS is the technical operator of the authentication software and responds to authentication requests. It provides the options for authenticating the Cardholder. The Issuer ACS is responsible for: • Determining the type of authentication needed based on the risk-based authentication rules set by the Issuer, • Authenticating a Cardholder according to the risk-based authentication rules set by the Issuer.

Interoperability domain

Component	Description
Directory Server (DS)	The DS is operated by the Payment Network and routes authentication messages between the Issuer and Acquirer. The DS is responsible for: • Routing messages between the appropriate 3DS Server and ACS, • Authenticating the ACS and 3DS Server, • Defining specific program rules (for example, logos, time-out values, etc.), • Onboarding 3DS Servers and ACS,

Component	Description
	- Maintaining ACS and DS protocol version lists and 3DS Method URLs, - Requesting the de-tokenization to the token service provider, - Sharing the list of eligible card data ranges to the 3DS Server.

Acquirer domain

Component	Description
3DS Requestor	The 3DS Requestor is the initiator of a 3DS authentication request. The Requestor may be a Merchant ecommerce platform, an application requesting the authentication as part of the checkout process. The 3DS Requestor is responsible for: - Providing a 3DS Client either via the 3DS Requestor App (3DS SDK), or the 3DS Method/Browser on the Consumer Device. The 3DS Requestor has a link or is integrated with the 3DS Server. - Initiating a 3DS authentication flow, directly through 3DS Server for a 3RI transaction, or by interfacing the 3DS Client with the 3DS Server. - Initiating authorization process upon authentication confirmation.
3DS Server (3DSS)	The 3DS Server is integrated within the 3DS Requestor environment to enable the communication between the 3DS Requestor and the Directory Server (DS). The 3DS Server is responsible for: - Handling the transactions, in particular starting the authentication, - Facilitating communication between 3DS Requestor and the DS, - Authenticating the DS, - Validating the DS, 3DS SDK, and the 3DS Requestor, - Ensuring the message content protection.

Component	Description
3DS Client	The 3DS Client enables consumer’s interactions with the 3DS Requestor for the initiation of the authentication. The transaction originates from the interactions between 3DS Client and 3DS Requestor. The 3DS Client can be Browser or an Application (3DS SDK). The 3DS Client is responsible for: i. The transaction initiation via 3DS Requestor interactions, ii. Enabling the challenges performed between the Issuer and the Cardholder for the authentication.

7.2.4 3DS message types

The table below show the different message types that are supported and exchanged between participants in JaywanDS.

Message	Message type	Frictionless	Challenge	Other
AReq/ARes	Authentication message	Yes	Yes	NA
CReq/CRes	Challenge message	NA	Yes	NA
RReq/RRes	Result message	NA	Yes	NA
PReq/PRes	Preparation message	NA	NA	Yes
Error	Error message	NA	NA	Yes

Authentication Request/Response (AReq/ARes)

The Authentication messages are present in both Frictionless and Challenge flow. The AReq message is the initial message in the 3DS authentication flow.

AReq	ARes
------	------

The 3DS Server forms the Areq message when requesting authentication of the Cardholder. It contains Cardholder, payment, and Device information for the transaction. The AReq message is sent by the 3DS Server to the DS which might update and redirects it to the ACS.	1. The ARes message is the ACS response to the AReq message. 2. It indicates that the Cardholder has been authenticated (Frictionless), or that further Cardholder interaction is required to complete the authentication (Challenge).
--	---

Challenge Request/Response (CReq/CRes)

The Challenge messages will be exchanged, if the Issuer ACS decides that strong Cardholder authentication is required.

CReq	CRes
The CReq message initiates Cardholder interaction in a Challenge Flow and used to carry authentication data from the Cardholder. 1. App-based: The CReq message is sent by the 3DS SDK. There are two or more CReq messages per challenge as multiple back-and-forth attempts between the ACS and the Cardholder may be required to complete the authentication. 2. Browser-based: The CReq message is sent by the 3DS Server. There is only one CReq message per challenge, but many back-	The CRes message is the ACS response to the CReq message. It indicates the result of the Cardholder authentication or, in the case of an App-based model, also signal that further Cardholder interaction is required to complete the authentication. 1. App-based: Elements of the CRes message provide the necessary data for the 3DS SDK to generate and display the user interface (UI) for the challenge. There are two or more CRes messages per transaction to complete Cardholder authentication.

and-forth messages using HTTP protocol (with HTML form).	2. Browser-based: The CRes message contains the authentication result and completes the Cardholder challenge. There is only one CRes message per challenge.
--	--

Result Request/Response (RReq/RRes)

The Result messages will be used to communicate the result of the challenge authentication issued by the issuer ACS.

RReq	RRes
1. The RReq message communicates the results of the authentication or verification. 2. The message is sent by the ACS through the DS to the 3DS Server. 3. There is per authentication flow only one RReq. 4. The RReq message is not present in a Frictionless transaction.	1. The RRes message acknowledges receipt of the RReq message. 2. The message is sent by the 3DS Server through the DS to the ACS. 3. There is only one RRes message per RReq message.

Preparation Request/Response (PReq/PRes)

The Preparation message is sent from the 3DS Server to the DS to request information about the ACSs and the DS. This message is not part of the 3DS authentication message flow.

PReq	PRes
------	------

1. PReq is sent from 3DS server to DS to request information about ACSs and DS. 2. The list of eligible card data ranges is contained in this response to PReq. and a 3DSS is supposed to request it every [1-24] hours to the DS.	1. The PRes message is the DS response to the PReq message. 2. The 3DS Server utilise the PRes message to cache information about the ACS and the DS (for example, about which Protocol Version(s) are supported).
--	--

Error Message

Error messages provide additional information about an error that occurred during message processing between the 3DS Server, the DS, the ACS, and the 3DS SDK.

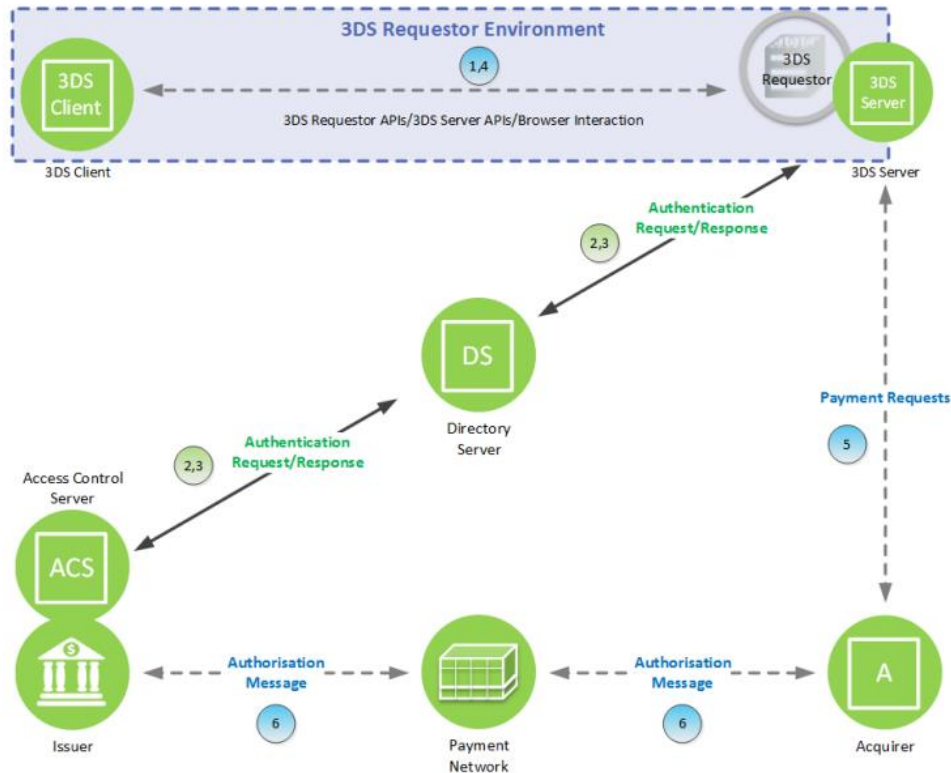
7.2.5 Authentication flows

JaywanDS support two authentication methods: Frictionless Authentication and Challenge Authentication.

Frictionless flow

The Frictionless Flow initiates a 3DS authentication flow and consists of an AReq message and an ARes message.

The Frictionless Flow does not require further Cardholder interaction to achieve a successful authentication and complete the 3DS authentication process.



1. 3DS Requestor environment: The necessary 3DS secure message is gathered and provided to the 3DS server. 3DS server includes all the data into AReq message. The information is provided based on the Device channel – App or browser based, Message category and 3DS Requestor 3D-Secure implementation.
2. Now the AReq is constructed at 3DS server and sent to DS, then the DS will forward the AReq to ACS.
3. In response to AReq message, the ACS will respond with ARes message to the DS, then DS forward the ARes message to 3DS server. Based on the information provided by AReq message, ACS will determine whether cardholder interaction is required or not required for authentication. It will also allow issuer to approve transaction without cardholder interaction.

-
4. 3DS server send the ARes message to the 3DS requestor environment which then inform the cardholder.

[Note: 3D secure process end here]

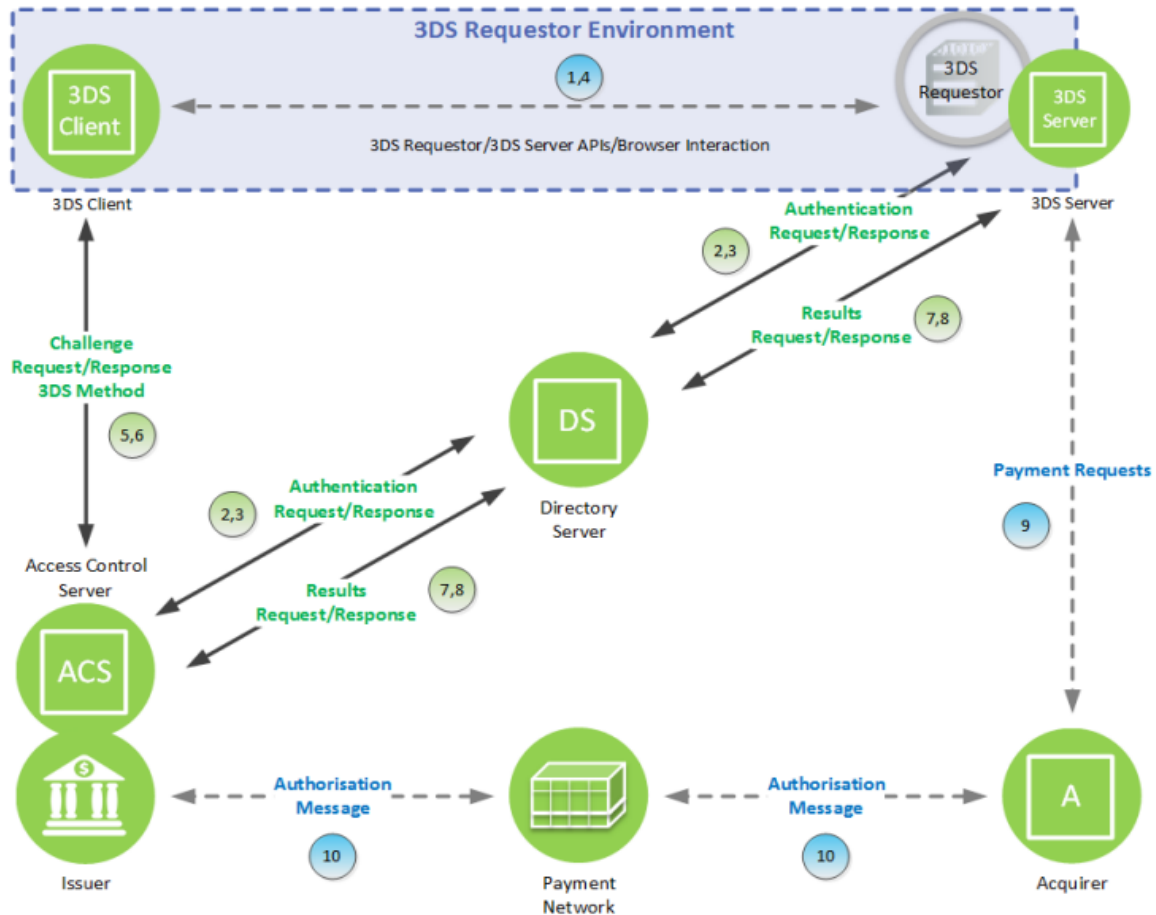
5. In step 5 merchant proceeds with authorization with its acquirer
6. The Acquirer now process authorization with issuer through payment gateway and after authorization returns the response message to merchant

Challenge flow

If the ACS determines that further Cardholder interaction is required to complete the authentication, the Frictionless Flow transitions into the Challenge Flow. For example, a challenge may be necessary because the transaction is deemed high-risk, is above certain thresholds, or requires a higher level of authentication due to country's regulatory mandates (or regulations).

In addition to the AReq and ARes messages that comprise the Frictionless flow, the Challenge flow completes with the RReq and RRes messages. The Challenge flow also includes CReq and CRes messages except in the case of Decoupled Authentication.

3DS Requestors decide whether to proceed with the challenge, or to terminate the 3DS authentication process.



Challenge flow consist of same steps followed by frictionless flow

1. 3DS Requestor environment: The necessary 3DS secure message is gathered and provided to the 3DS server. 3DS server includes all the data into AReq message. The information is provided based on the Device channel – App or browser based, Message category and 3DS Requestor 3D-Secure implementation.
 2. Now the AReq is constructed at 3DS server and sent to DS, then the DS will forward the AReq to ACS.
 3. In response to AReq message, the ACS will respond with ARes message to the DS, then DS forward the ARes message to 3DS server. Based on the information provided by AReq message, ACS will determine whether cardholder interaction is required or not required for authentication.
- So for challenge flow cardholder information is required to complete the authentication

4. 3DS server send the ARes message to the 3DS requestor environment which then inform the cardholder.
5. 3DS client initiated the CReq message based on n information received in the ARes message. The manner in which this is done depends on the model:
 - App-based—A CReq message is formed by the 3DS SDK and is posted to the ACS URL received from the ARes message.
 - Browser-based—A CReq message is formed by the 3DS Server and is posted through the Cardholder's browser by the 3DS Requestor to the ACS URL received from the ARes message
6. ACS received the CReq message from 3DS client for cardholder authentication.
 - App-based—The ACS utilises pairs of CReq and CRes messages to perform the challenge. In response to the CReq message, the CRes message requesting the Cardholder to enter the authentication data is formed by the ACS and sent to the 3DS SDK.
 - Browser-based—The ACS sends the authentication user interface to the Cardholder browser. The Cardholder enters the authentication data via the browser to be checked by the ACS. In response to the CReq message, the CRes message is formed by the ACS and sent to the 3DS Server to indicate the result of the authentication

[**Note:** For the App-based model, Step 5 and Step 6 will be repeated until the ACS makes a determination.]
7. The ACS send as RReq message to DS including Authentication value (AV), then sends the message to 3DS server using 3DS server URL received from AReq message.
8. The 3DS server responds to the received RReq message to the DS with ARes and then DS routes it to the ACS.

7.2.6 Authentication channels

The table below shows the JaywanDS supported authentication channels:

Channel	Definition
App-based (01-APP)	1. App-based channel refers to the authentication channel where a Merchant application is used, integrating the 3DS Client. 2. The Merchant application integrates a 3DS Software Development Kit (3DS SDK) that enables the link between the Merchant application and JaywanDS 3DS modules (3DS Server & ACS).
Browser-based (02-BRW)	1. Browser-based channel refers to the authentication channel where the 3DS Client is integrated within the Merchant's website and invoked using the browser on the consumer device. 2. The 3DS Requestor, utilises the browser's embedded feature such as JavaScript to gather relevant authentication data, and HTML engine to display the UI of a cardholder authentication challenge. 3. The communication flow between the Consumer Device and the 3DS Server/3DS Requestor uses HTTP over TLS.
3DS Requestor Initiated (03-3RI)	1. The 3RI channel is used to perform deviceless payment authentication or account verification. 2. The authentication flow is initiated by the merchant without Cardholder's presence and are processed by back-end systems, therefore there are no processing screens, or 3DS Client involved, outside the required OOB, Decoupled Authentication (refer to [EMV-3DS-CORE-SPECIFICATION] for details). 3. 3RI transactions are also referred to as Merchant Initiated Transaction (MIT).

7.2.7 Transaction Status Indicators

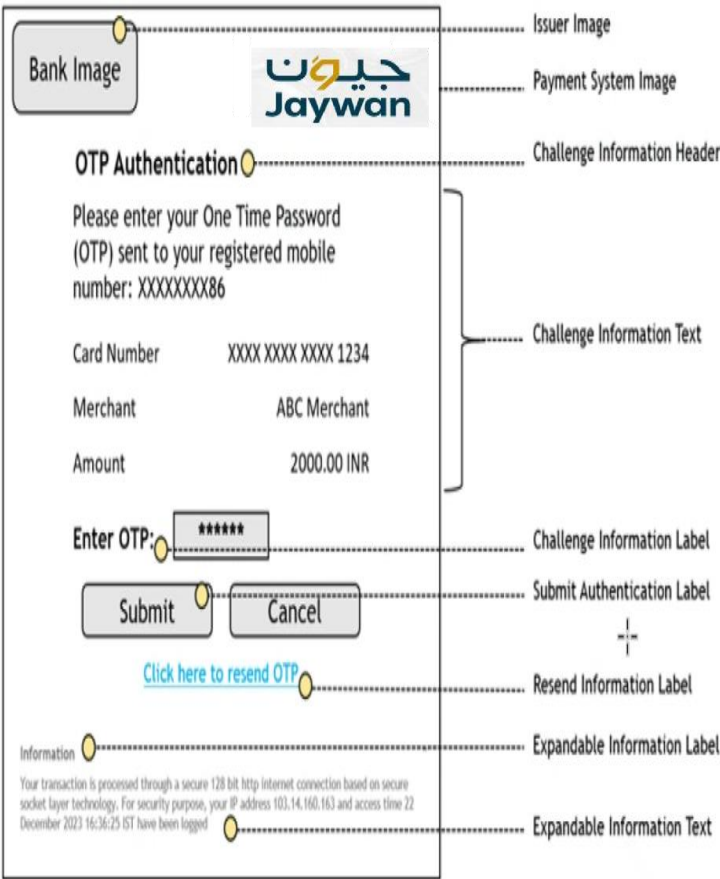
The Transaction Status (transStatus) indicates whether a transaction qualifies as an authenticated transaction or account verification.

Indicator	Description
Y	Authentication Verification Successful
N	Not Authenticated/Account Not Verified; Transaction denied
U	Authentication/Account Verification Could Not Be Performed; Technical or other problem, as indicated in the ARes or RReq
A	Attempts Processing Performed; Not Authenticated/Verified, but a proof of attempted authentication/verification is provided
C	Challenge Required; Additional authentication is required using the CReq/CRes
D	Challenge Required; Decoupled Authentication confirmed
R	Authentication/ Account Verification Rejected; Issuer is rejecting authentication/verification and request that authorisation not be attempted.
I	Informational Only; 3DS Requestor challenge preference acknowledged

7.2.8 Checkout UI template

To ensure a consistent user experience across both App-based (Native or HTML) and Browser-based challenge interactions, JaywanDS establishes a standardized UI template derived from the EMV 3-D Secure UI Template. The initial CRes message sent by the ACS contains essential data elements, which are subsequently employed to populate the challenge interface, as illustrated below.

The JaywanDS UI Template incorporates specific EMV 3-D Secure data elements, outlined in the table below. These elements are intricately connected with the corresponding interface sample presented thereafter, ensuring cohesion and consistency in the user interaction flow.



The following table links the data element of the UI Template to their related EMV 3-D Secure fields.

Data Element	Field	Value	
Issuer Image	issuerImage	param	Value
		medium	Fully qualified URL to medium resolution Issuer's logo
		high	Fully qualified URL to high resolution Issuer's logo
		extraHigh	Fully qualified URL to extra high-resolution Issuer's logo
		Example <pre>"issuerImage":{ "medium":"http://is.com/m_image.png", "high":"http://is.com/h_image.png", "extraHigh":"http://is.com/e_image.png" }</pre>	
Payment System Image	psImage	param	Value
		medium	Fully qualified URL to medium resolution Network logo
		high	Fully qualified URL to high resolution Network logo

Data Element	Field	Value		
		<table><tr><td>extraHigh</td><td>Fully qualified URL to extra high-resolution Network logo</td></tr></table> Example <pre>"psImage":{ "medium":"http://ds.com/m_image.png", "high":"http://ds.com/h_image.png", "extraHigh":"http://ds.com/e_image.png" }</pre>	extraHigh	Fully qualified URL to extra high-resolution Network logo
extraHigh	Fully qualified URL to extra high-resolution Network logo			
Challenge Information Header	challengeInfoHeader	Header text of the challenge information. Should contain relevant information realated to the type of challenge performed. Example <pre>"OTP Authentication"</pre>		
Challenge Information Text	challengeInfoText	Detailed information of the challenge type and key information about the transaction. The information presented should enable the cardholder to identify the transaction. Example <pre>"Please enter your One Time Password (OTP) sent to your registered mobile number: XXXXXXXX86\n \n Card Number\tXXXX XXXX XXXX 1234\n Merchant\tABC Merchant\n Amount\t2000.00 INR"</pre>		

Data Element	Field	Value
Challenge Information Label	challengeInfoLabel	Label of the challenge data entry. Should be related to the type of challenge performed. Example "Enter OTP"
Submit Authentication Label	submitAuthenticationLabel	Label of the submit challenge data button. Example "Submit"
Resend Authentication Label	resendInformationLabel	Label of the button to request the ACS to resend the challenge information. Example "Click here to resend OTP"
Expandable Information Label	expandInfoLabel	Label of the information section. Example "Information"
Expandable Information Text	expandInfoText	Information about the authentication method provided by the Issuer. Example "Your transaction is processed through a secure 128-bit http internet connection based on secure socket layer technology."

7.3 CAVV computation

7.3.1 CAVV generation

Assembling authentication value.

Position	Length	Field Name	Data	Byte Number
1	1 byte (2BCD)	3-D Secure Authentication Results Code	"00" – Successful Authentication (status Y) "05" – Authentication could not be performed (status U) "07" – Attempt Processed by ICS ACS "08" – Attempt Processed by ICS DS "09" – Authentication failed (status N)	1
2	1 byte (2BCD)	Second Factor Authentication Results Code	▪ 01 = Static Passcode ▪ 02 = SMS OTP ▪ 03 = Key fob or EMV card reader OTP ▪ 04 = App OTP ▪ 05 = OTP Other ▪ 06 = KBA ▪ 07 = OOB Biometrics ▪ 08 = OOB Login ▪ 09 = OOB Other ▪ 10 = Other ▪ 11 = Push Confirmation ▪ 12 = Decoupled ▪ 98 = EMV 3DS Attempt ▪ 99 = Frictionless	2

Position	Length	Field Name	Data	Byte Number
3	1 byte (2BCD)	AVV Key indicator	01: Issuer CAVV Key Prod 02: Issuer CAVV Key Test 10: DS CAVV Key Prod 11: DS CAVV Key Test	3
4-5	2 bytes (4BCD)	CAVV output	CAVV Output computed by ACS or DS A leading zero is required to pad the first unused byte of the CAVV. Refer below section for calculation.	4-5
6-7	2 bytes (4BCD)	Unpredictable Number	Last 4 digits of ATN.	6-7
8-15	8 bytes (16BCD)	ATN	16-digit transaction identifier generated by the ACS	8-15
16-17	2 bytes (4BCD)	Reserved	Zero filled	16-17
18-20	3 bytes (6BCD)	Date	YYDDD	18-20

The above 20 bytes value should be base64 encoded and passed in authenticationValue field of ARes/RReq response.

7.3.2 CAVV calculation

The CAVV encryption shared key is used to generate the CAV Output and is composed of two keys, Key A and Key B. The ACS shall generate the CAV Output as follows.

Input for CVV2 calculation	
Field Name	Length
PAN	13 to 19
ATN	4

(Last 4 digits of ATN)	
Authentication Results Code (lower half of the 2BCD)	1
Second Factor Authentication Results Code	2

- A. Concatenate the data in the above table
- B. Fill the data created in (A) with zeros up to 128 bits.
- C. Divide the data from (B) into two blocks of 64 bits each.
- D. DES encrypt the first half-block with Key A.
- E. XOR the result from (D) with the second half block and DES encrypt the result with Key A.
- F. DES decrypt the result from (E) with Key B.
- G. DES encrypt the result from (F) with Key A.
- H. Extract 0-9 values from the result from (G).
- I. Extract A-F values from the result from (G) and replace them with the value after subtracting 10 from them.
- J. Concatenate (I) onto the end of (H).
- K. Use the upper three digits of (J) as the CAV Output

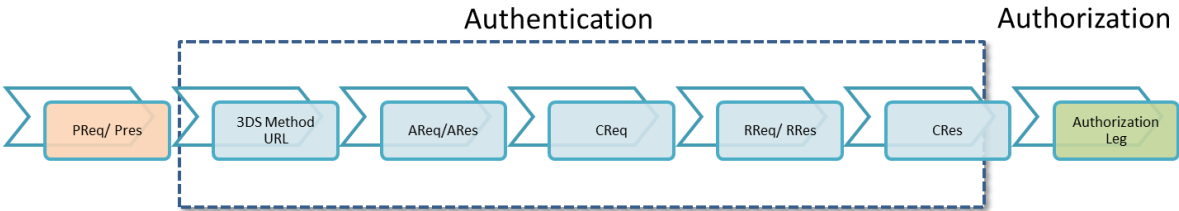
7.3.3 Authentication Result Code Derivation

CAVV Data Field	
Data Field	Sample Value
PAN	6522 2967 8901 2344
ATN (Last 4 digits of ATN)	8735
Authentication Results Code	5

(lower half of the 2BCD)	
Second Factor Authentication Results Code	01
CAVV Value Calculation Examples	
Step	Example (using Sample Values above)
Extract data	6522 2967 8901 2344 8735 5 01
Place into 128-bit Field	6522 2967 8901 2344 8735 5 01 0000000000
Split into 2 64-bit blocks	Block 1 = '6522 2967 8901 2344' Block 2 = '8735 5010 0000 0000'
Using DES, encrypt Block 1 using Key A	Block 1 = '6522 2967 8901 2344' Key A = "0123 4567 89AB CDEF" Result = "D4D5 E07D 03C8 B0A4"
XOR result of Step 4 w Block 2, then encrypt the XOR result with Key A	Block 2 = "8735 5010 0000 0000" XOR Result = "53E0 B06D 03C8 B0A4" Key A = "0123 4567 89AB CDEF" Result = " DA00 F1F7 DE86 E304"
Using DES, decrypt the result of Step 5 with Key B	Key B = "FEDC BA98 7654 3210" Result = " 66B6 71A2 9FBC B2E5"
Encrypt the result of Step 6 with Key A	Key A = "0123 4567 89AB CDEF" Result = "3D3F5F209414B1AE "
Extract all digits	Result = "3352 0941 41"
Extract hex digits from Step 7 and convert to numeric (subtract 10 from each)	Hex Digits = "DFFB AE" Result = "3551 04"
Concatenate results of Step 9 and Step 8	Result = "3352 0941 4135 5104"
Select 3 left-most digits. This is the CVV2 output	Result = "335"

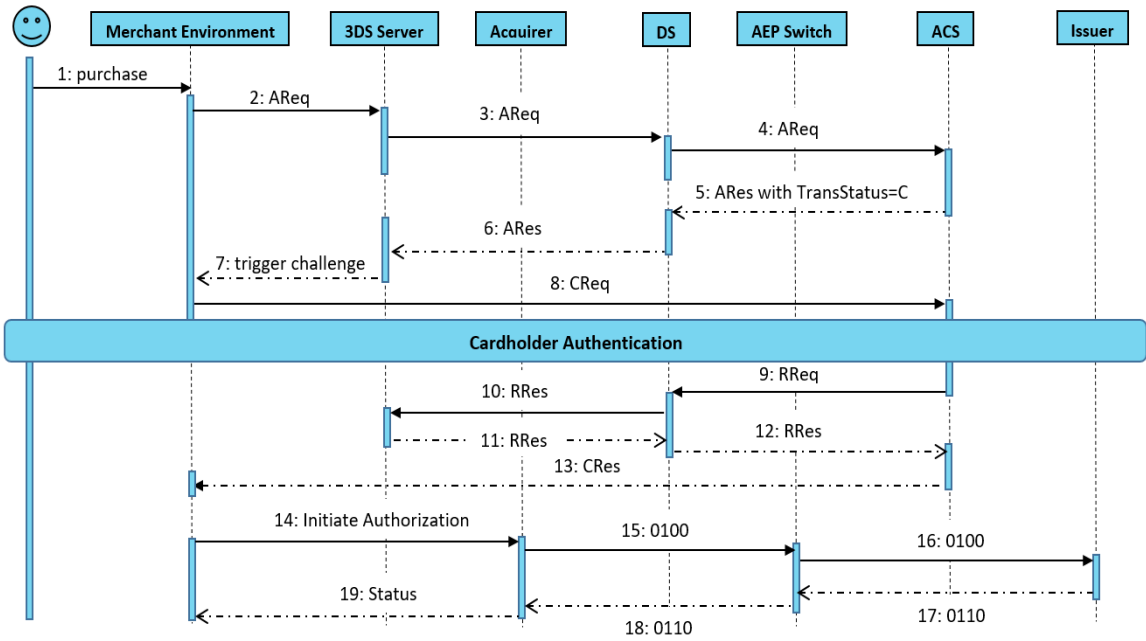
8 Transaction Flow

Following is the flow of 3DS message type in the 3DS transaction Journey for payment transactions.



Following is the transaction journey of the ECOM 3DS transactions in JaywanDS.

Sequence Diagram



Detailed sequence

1. The cardholder purchases a product and proceed to checkout on merchant page.
 - a. The merchant initiates the 3DS Method URL and complete the process as describe in **section below Chapter 7.2 - 3DS Method**
2. The merchant initiates the authentication message AReq.

3. The AReq is sent by the 3DS Server to the DS.
4. The AReq is transmitted by the DS to the ACS.
5. The ACS returns ARes with *transStatus=C* to trigger a challenge.
6. The DS forwards ARes to the 3DS Server.
7. The 3DS Server notifies the merchant environment about the status.
8. The merchant environment triggers the CReq. An html form is submitted within an iframe on the merchant web page. The html form contains hidden input containing the base64url encoded CReq provided by the 3DS Server.

Cardholder Authentication

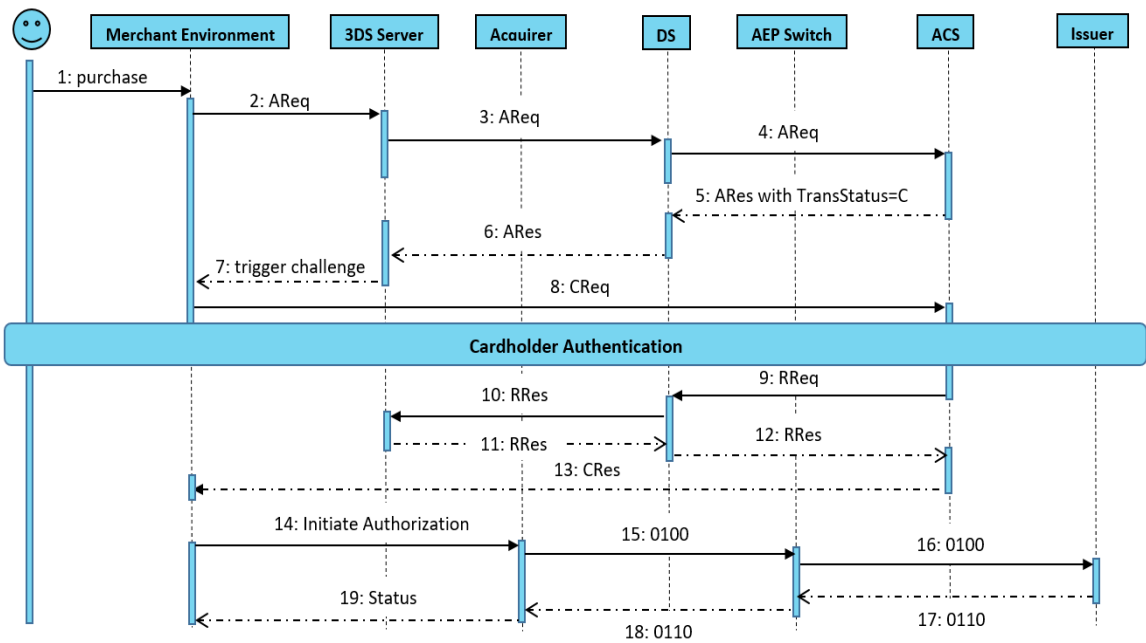
- a. Within the iframe, the ACS returns an html/css/js page displaying information to the cardholder.
 - b. The cardholder fulfils the requested information (entering OTP, passcode, etc.).
 - c. The ACS html form with the cardholder data is submitted to the ACS.
9. The ACS sends the RReq with the authentication result to the DS.
 10. The DS forwards the RReq to the 3DS Server.
 11. The 3DS Server returns the RRes to the DS.
 12. The DS forwards the RRes to the ACS.
 13. The ACS sends the final CRes to the merchant environment.
 14. The merchant initiates the authorization.
 15. The acquirer sends the *authorization (0100)* to the Payment Network switch.
 16. The Payment Network switch forwards the message to the issuer host.
 17. The issuer host replies with the response *(0110)*.
 18. The response is forwarded to the acquirer.
 19. The acquirer shares the status to the merchant environment.

8.1 Payment Transaction.

The cardholder initiates a payment transaction on a merchant website or application to purchase goods or services.

8.1.1 JaywanDS - Challenge Authentication Purchase Transaction.

Sequence Diagram



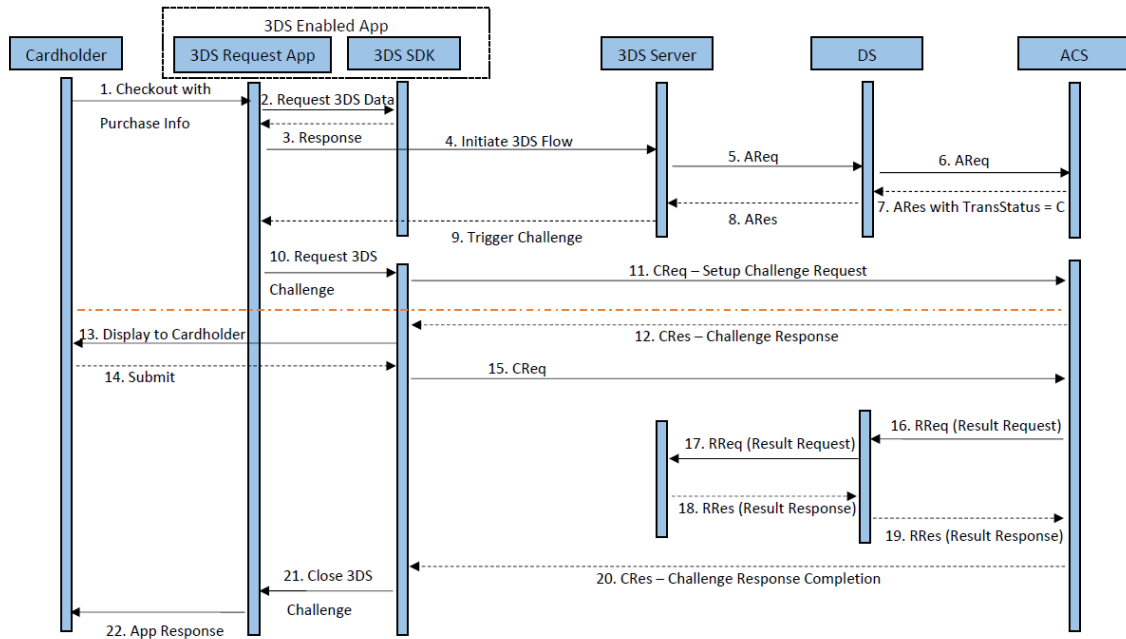
Detailed sequence

1. The cardholder initiates the authentication for the purchase transaction.
2. The merchant initiates the Authentication Request to 3DSS.
3. The 3DSS sends the AReq to the JaywanDS.
4. The JaywanDS forwards the AReq to the ACS. The ACS responds to AReq with a challenge required response.
5. The ACS sends the ARes to the JaywanDS

-
6. The JaywanDS forwards the ARes to the 3DS Server.
 7. The 3DS requestor environment triggers the challenge flow.
 8. The first CReq is sent to the ACS
 9. The ACS validates the challenge and creates the RReq message. If the user has been successfully authenticated, the issuer generates a CAVV for the total amount. The RReq is sent to the DS.
 10. The JaywanDS forwards the RReq to the 3DSS.
 11. The 3DSS creates and sends the RRes to the JaywanDS.
 12. The JaywanDS sends the RRes to the ACS.
 13. The ACS sends the final CRes to the merchant environment.
 14. The merchant initiates the authorization to the acquirer switch.
 15. The acquirer Switch sends the authorization E-Commerce 3D Purchase to the Payment Network switch.
 16. The Payment Network switch forwards the message to the issuer switch.
 17. The issuer switch replies with the response 0110
 18. The response is forwarded to the acquirer.
 19. The acquirer shares the status to the merchant environment.

8.1.2 JaywanDS – Challenge Authentication Purchase Transaction (3DS SDK Flow)

Sequence Diagram



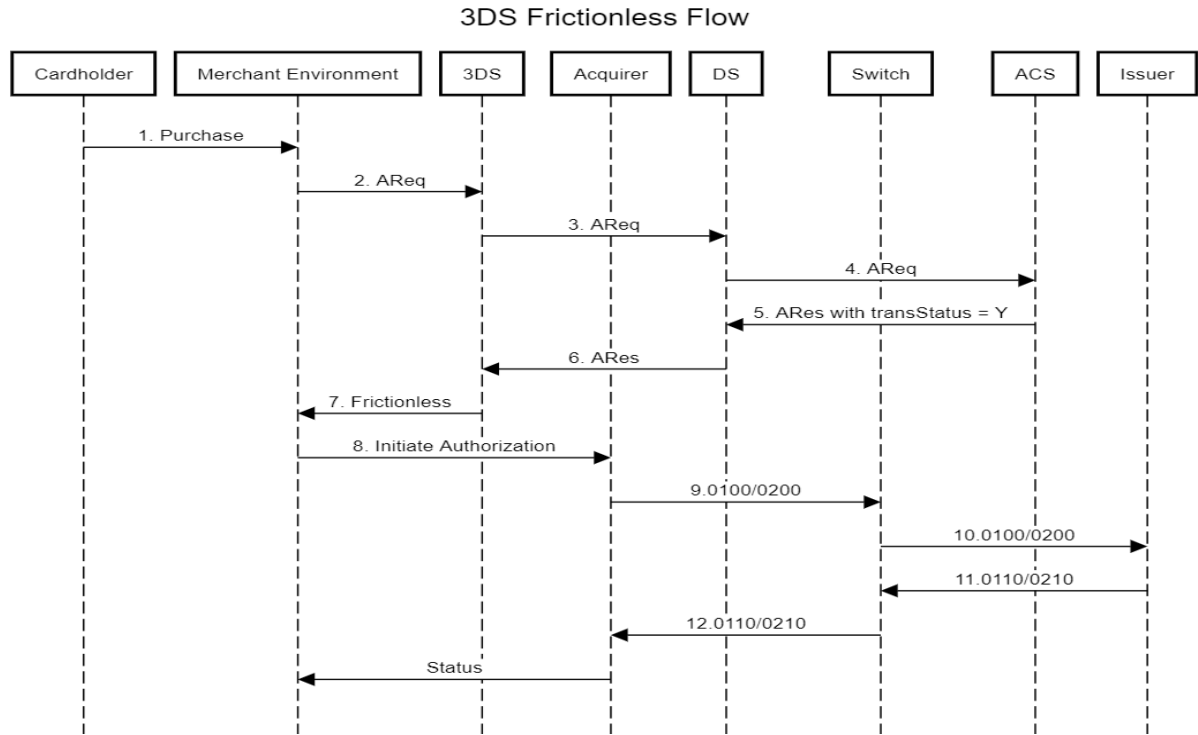
1. The cardholder initiates the authentication for the purchase transaction.
2. The 3DS Rquestor App requests 3DS data like SDK Transaction ID, SDK ID etc from 3DS SDK.
3. The 3DS SDK sends the response back to Requestor App.
4. The Requestor App requests 3DS Server to initiate the flow.
5. 3DS Server forms the AReq basis the detail received and sends to JaywanDS.
6. DS decrypts the sdkencdata field and extract details. Forms the AReq message and sends to ACS.
7. The ACS sends the ARes to the JaywanDS with TransStatus = C for challenge.
8. The JaywanDS DS forwards the ARes to the 3DS Server.
9. The 3DS Server sends response back to Requestor App environment to trigger the challenge flow.

-
10. 3DS Requestor gathers the necessary ARes data elements and make it available to 3DS SDK to request the 3DS Challenge.
 11. 3DS SDK completes the secure channel setup with ACS and sends the first CReq (challenge request) to ACS.
 12. ACS validates the sender and forms the challenge and send to 3DS SDK in CRes (challenge response).

There could be multiple CReq/CRes communication between 3DS SDK & ACS as per the challenge requirement.
 13. 3DS SDK displays the challenge UI to cardholder.
 14. Cardholder submits the challenge.
 15. 3DS SDK sends the challenge request with submitted data to ACS.
 16. The ACS validates the challenge and creates the RReq message. If the cardholder has been successfully authenticated, the issuer generates a CAVV. The RReq is sent to the JaywanDS.
 17. JaywanDS sends the Authentication Request (AReq) to 3DS Server informing about authentication status.
 18. The 3DSS creates and sends the RRes to the JaywanDS.
 19. The JaywanDS sends the RRes to the ACS.
 20. The ACS sends the final CRes to the 3DS SDK.
 21. 3DS SDK closes the challenge and inform 3DS Requestor App.
- 3DS Requestor App shows the final response to cardholder.

8.1.3 JaywanDS – FrictionLess Purchase Transaction.

Sequence Diagram



Detailed sequence

1. The cardholder initiates the authentication for the purchase transaction.
2. The merchant initiates the Authentication Request to 3DSS.
3. The 3DSS sends the AReq to the JaywanDS.
4. The JaywanDS DS forwards the AReq to the ACS. The ACS triggers a frictionless.
5. The ACS sends the ARes to the JaywanDS DS with transStatus as Y
6. The JaywanDS DS forwards the ARes to the 3DS Server.
7. The 3DS requestor environment triggers the Frictionless flow.
8. The acquirer sends the authorization E-Commerce 3D Purchase to the Payment Network Switch.
9. The CB-UAE switch forwards the message to the issuer switch.
10. The issuer switch replies with the response 0110
11. The response is forwarded to the acquirer.

12. The acquirer shares the status to the merchant environment

8.2 3DS method

The 3DS method is a mechanism defined in **[EMV-3DS-CORE-SPECIFICATION]** allowing the issuer to have direct access to the cardholder’s browser on the merchant web site.

- 1. Only available for browser-based flow, it permits to improve device recognition and therefore to increase approval rate.
- 2. **Req 402** The inclusion of *3DS Method URL* and account ranges in the JaywanDS DS is recommended.
- 3. The 3DS Requestor and 3DS Server shall rely on the cached PRes message data to determine the ACS 3DS Method URL via the Card Range Data data element.
- 4. **Req 404** The 3DS Requestor SHOULD use 3DS method to improve the conversion rate. Although strongly recommended, it is not mandatory.

Subject	Description
Time-out	Req 402 Even if [EMV-3DS-CORE-SPECIFICATION] allows up to 10 sec, JaywanDS limits the process to 5 sec.
Browser fingerprint	The way to gather device information is ACS specific. Industry best practice is to combine several technics to increase accuracy, such as ▪ HTML5 canvas ▪ CSS rendering analysing ▪ WebGL based analysis ○ Canvas ○ Audio ○ hardware ▪ browser plugin list

Web privacy	ACS SHOULD consider web privacy trends such as user agent reduction [UA CLINT], third party cookie removal and IP blindness.
Browser capacity	Browser capacity can be detected in order to detect efficient authentication method if needed, and to prevent lack of information from merchant. Browser capacities such as, <u>but not limited to</u>: ▪ JS enabled ▪ Screen properties ▪ User agent ▪ Local settings ▪ FIDO2 support (WebAuthn and SPC)

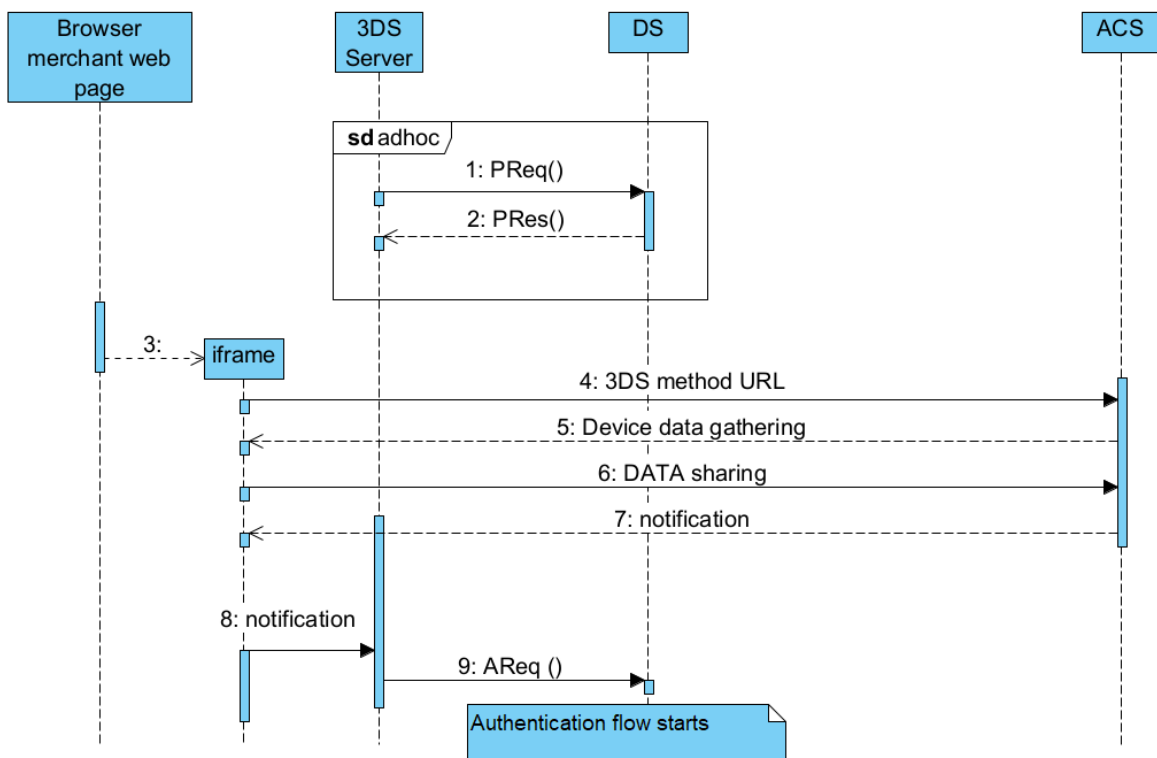


Figure 1 3DS method sequence flow

Sequence Diagram

Detailed sequence

1. On regular basis, the 3DS Server fetches ACS card range data from the JaywanDS DS. This is done by sending the preparation message PReq as defined in [\[EMV-3DS-CORE-SPECIFICATION\]](#)

2. **Req 402** The 3DS method URL SHOULD be included in the PRes response.
3. To 4. On checkout, the merchant calls the 3DS method URL on ACS side. The way the 3DS method URL is transmitted to the merchant is proprietary.

The merchant prepares a HTML form with following values

Form input	value			
Type=hidden	Base64url encoded value of			
Name=threeDSMethodData	<table><tr><td>Param</td></tr><tr><td>threeDSServerTransID</td></tr><tr><td>threeDSMethodNotificationURL</td></tr></table>	Param	threeDSServerTransID	threeDSMethodNotificationURL
Param				
threeDSServerTransID				
threeDSMethodNotificationURL				

Example

```
<form name="frm" method="POST" action="3DSMethodURL">
  <input type="hidden" name="threeDSMethodData"
    value="eyJ0aHJIZURTU2VydmVyVHJhbnNJRCI6IjNhYzdjYWE3LWFhNDItMjY2MjY0OTFiLTJhYzA1YTU0MmM0YSIsInRocmVIRFNNZXRob2ROb3RpZmljYXRpb25VUkwiOiJodHRwczovLzNkc21ldGhvZG5vdGlmaWNhdGlubi4zZHNzLmluIn0">
</form>
```

With threeDSMethodData decoded value

```
{
  "threeDSServerTransID":"3ac7caa7-aa42-2663-791b-2ac05a542c4a",
  "threeDSMethodNotificationURL":"https://3dsmethodnotification.3dss.in"
}
```

5. In return the ACS sends javascript code to gather device information such as browser fingerprinting.
6. The ACS javascript sends information to the ACS endpoint

7. Once done, the ACS notifies the 3DS Server about the 3DS method completion by submitting a form to the 3DS server on notification URL.

The ACS prepares a HTML form with following values

Form input	value		
Type=hidden	Base64url encoded value of		
Name=threeDSMethodData	<table><tr><th>Param</th></tr><tr><td>threeDSServerTransID</td></tr></table>	Param	threeDSServerTransID
Param			
threeDSServerTransID			

Example

```
<form name="frm" method="POST" action="threeDSMethod NotificationURL">
  <input type="hidden" name="threeDSMethodData"
    value="
eyJ0aHJlZURTU2VydmVyVHJhbnNJRCI6ljNhYzdjYWE3LWFhNDItMjY2My03OTFiLTJhYzA1YTU0MmM0YSJ9">
</form>
```

With threeDSMethodData decoded value

```
{
  "threeDSServerTransID":"3ac7caa7-aa42-2663-791b-
2ac05a542c4a"
}
```

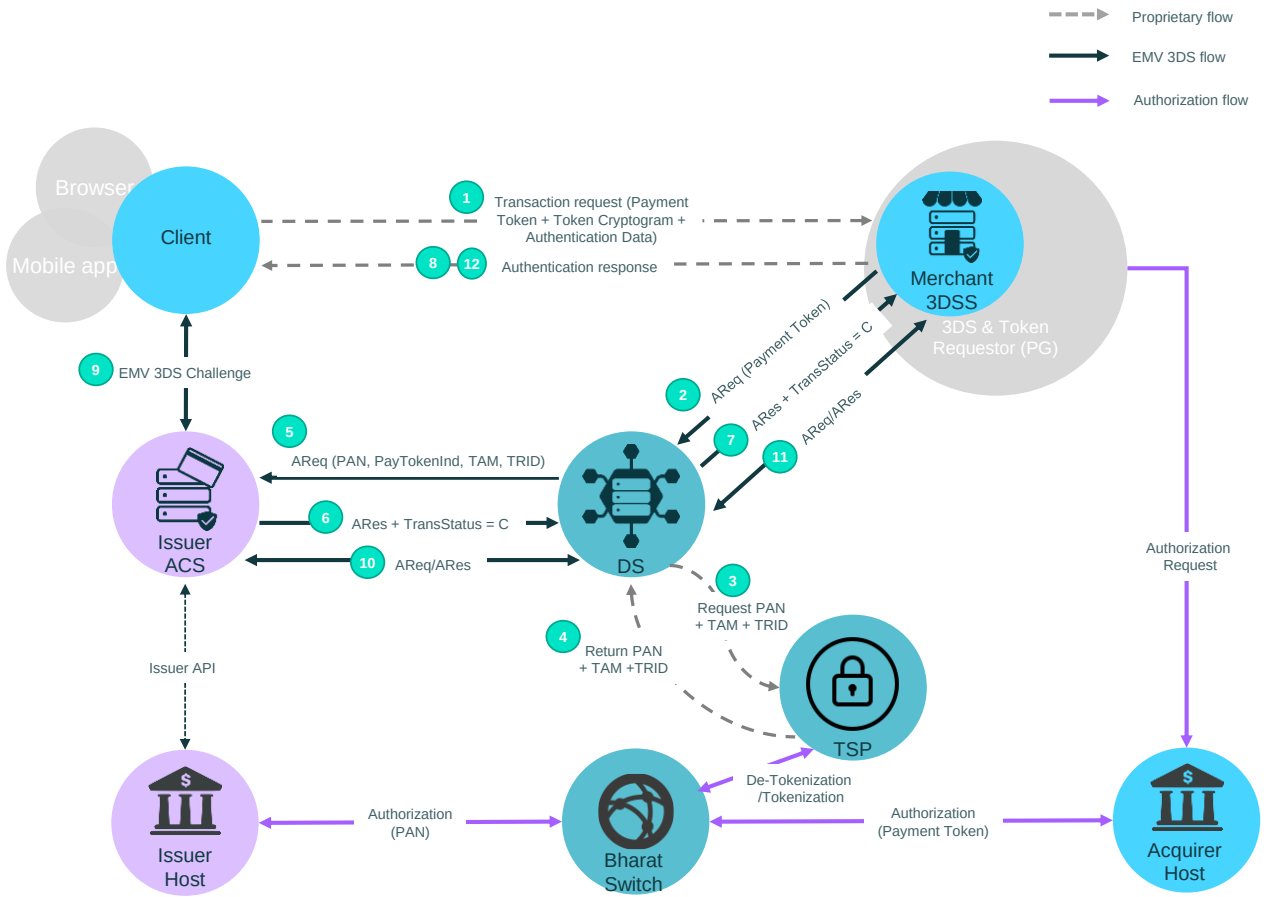
8. The form is submitted to the 3DS Server threeDS Method Notification URL
9. The 3DS Server initiates the authentication flow (AReq) and set the threeDSComplInd field accordingly

Field	Value
threeDSComplInd	Y = Successfully completed N = Did not successfully complete U = Unavailable—3DS Method URL was not present in the PRes message data for the card range associated with the Cardholder Account Number.

9 Tokenization

The payment ecosystem is evolving to support payment form factors that provide increased protection against counterfeit, account misuse, and other forms of fraud. Tokenisation is the process of replacing a consumer’s PAN with a unique, payment token, it’s offers consumers and merchants the ability to protect payment data throughout a transaction to support security. This enhances the underlying security of digital and online payments by limiting the risk of the PAN being compromised or used fraudulently / without authorisation. It also reduces potential disruption for both consumers and merchants if a payment card is lost or stolen.

JaywanDS program relies on NTS module to process the tokenization / de-tokenization of a token that will be used in the EMV 3-D Secure authentication process with JaywanDS DS. As EMV 3-D Secure protocol v2.2.0 support message extensions that have been defined in order to support EMV payment tokenization. It has been possible to adapt the usage of the message extension to the NTS needs.



9.1 Tokenized Payment Flow.

Card-on-file is when the cardholder's consent, stores the payment card data in the merchant secure wallet. The cardholder and merchant can then reuse the details for future payments and faster checkouts. As mandated by Regulators regulations, the merchant is not allowed to store directly the card data (such as the PAN and PAN expiry date), therefore the tokenization is required for Card-on-file use cases.

This section defines transaction flow for card on file using Tokenization.

9.1.1 JaywanDS – Tokenized Challenge Flow.

Sequence diagram

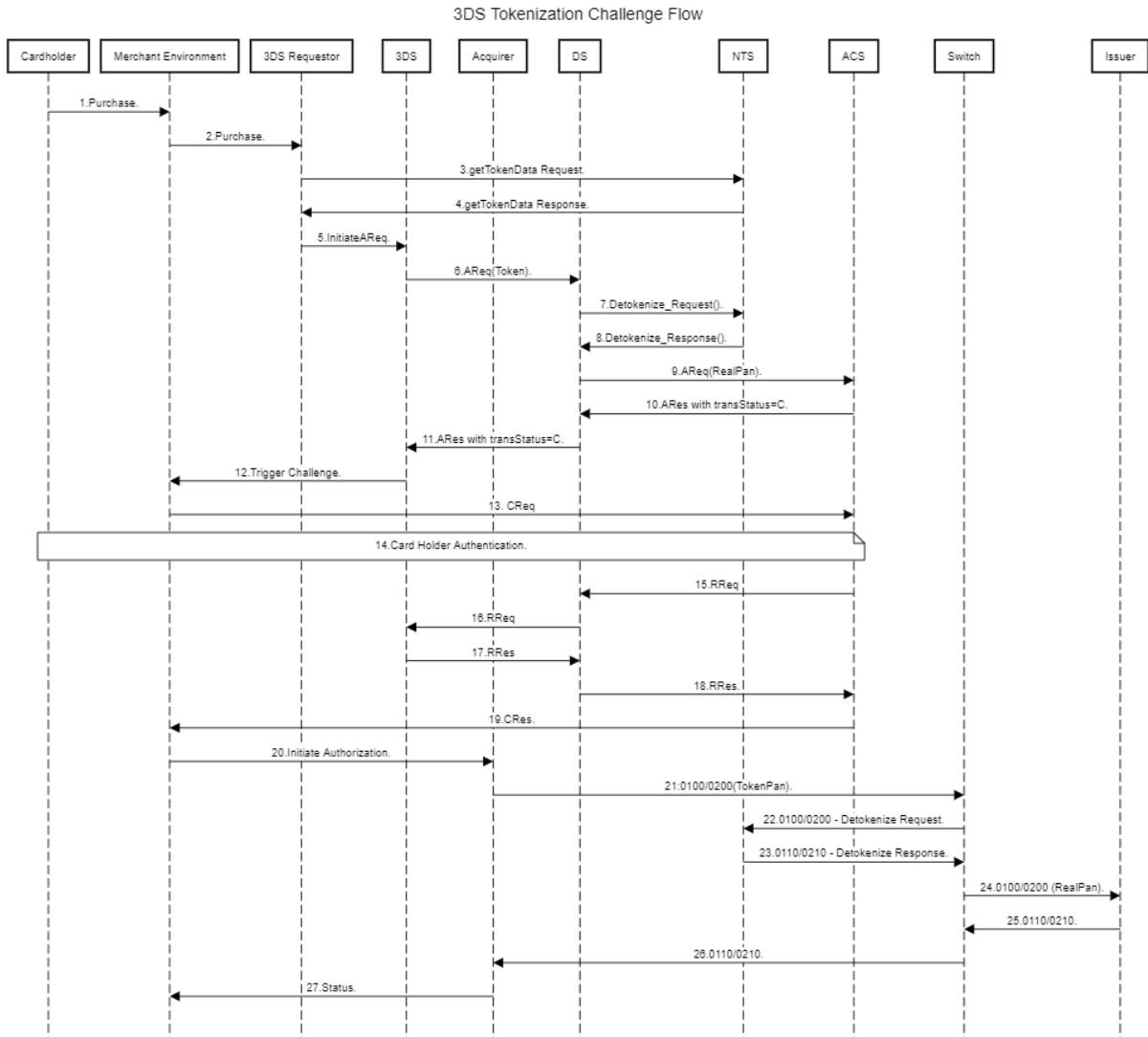


Figure 2 JaywanDS – COFT - Add Card

Detailed sequence

1. The cardholder purchases a good/service with a card previously added.
2. The merchant initiates the purchase
3. The Token Requestor initiates a getTokenData () through the get_token API to NTS
4. NTS will respond with the Token data (TokenPan, Expiry and TAVV).

5. The Merchant 3DS/Token Requestor Initiates the Areq for first purchase
6. The 3DS Server generates the AReq as described in [EMV-3DS-CORE-2.2.0] and sends the message to the JaywanDS DS.
7. The JaywanDS DS sends a request for detokenize Request () to NTS, to validate the token data
8. NTS will respond with realPan and expiry date.
9. The JaywanDS DS forwards the AReq by replacing the token and token related data by the PAN and card data to the Issuer ACS as follow.
10. The ACS sends the ARes to the JaywanDS DS with transStatus=C
11. The JaywanDS DS forwards the ARes to the 3DS Server.
12. The 3DS requestor environment triggers the Challenge flow.
13. The first CReq is sent to the ACS.
14. Customer perform authentication.
15. The ACS validates the challenge and creates the RReq message. If the user has been successfully authenticated, the issuer generates a CAVV for the total amount. The RReq is sent to the DS.
16. The JaywanDS DS forwards the RReq to the 3DSS.
17. The 3DSS creates and sends the RRes to the DS.
18. The JaywanDS DS sends the RRes to the ACS.
19. The ACS sends the final CRes to the merchant environment.
20. The merchant initiates the authorization flow to the Acquirer.
21. The Acquirer sends the authorization message with the authentication value. Among all ISO 8583 fields to Payment Network switch.
22. The switch requests a detokenization to NTS to retrieve the PAN.
23. NTS will respond with RealPan and ExpiryDate details.
24. The CB-UAE switch update the Primary Account Number and Date, Expiry in the ISO 8583 message before to send it to the issuer host.
25. The issuer host creates ISO 8583 response to Payment Network Switch

26. The Payment Network Switch sends the ISO 8583 response to the acquirer
27. The acquirer system returns the transaction status to the merchant (Token Requestor)

9.1.2 JaywanDS - Tokenized Frictionless Flow.

Sequence diagram

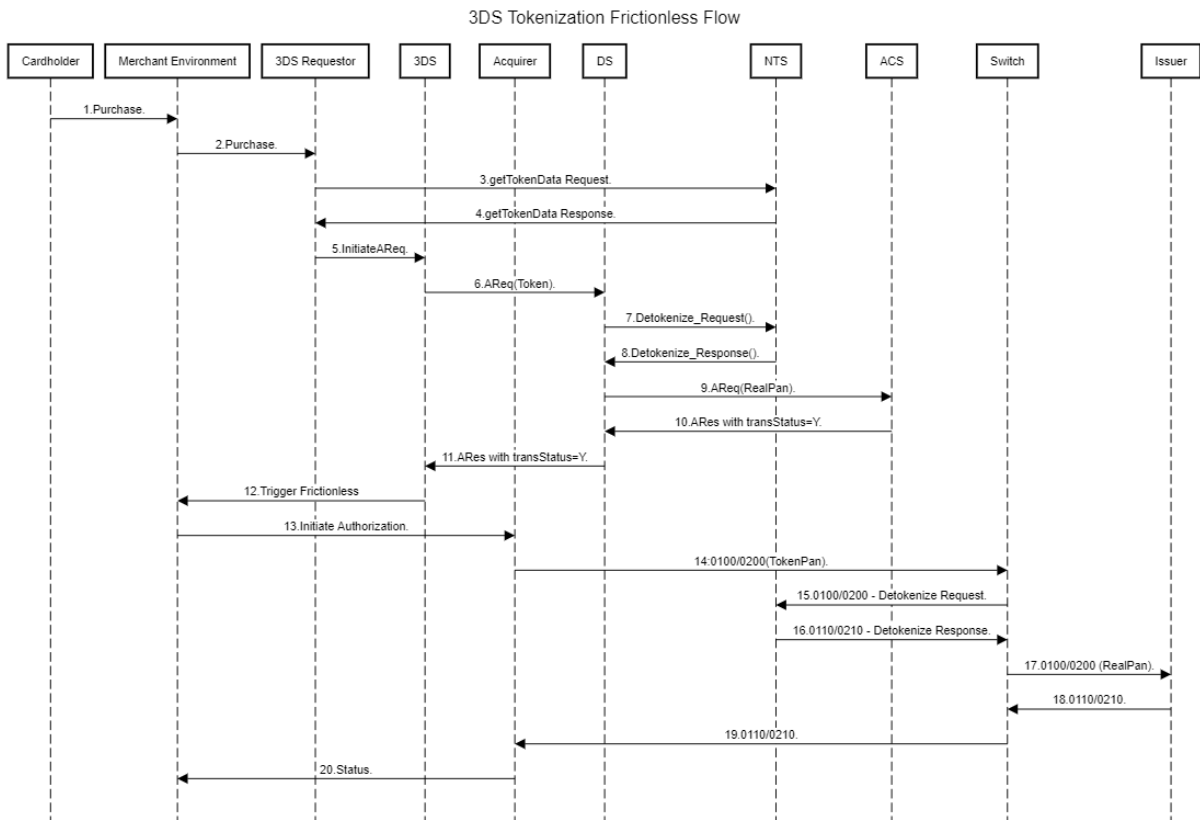


Figure 3 JaywanDS – COFT - Add card inline with a Regular Payment

Detailed sequence

1. The cardholder purchases a good/service with a card previously added.
2. The merchant initiates the purchase
3. The Token Requestor initiates a getTokenData () through the get_token API to NTS
4. NTS will respond with the Token data (TokenPan, Expiry and TAVV).
5. The Merchant 3DS/Token Requestor Initiates the AReq for first purchase

6. The 3DS Server generates the AReq as described in [EMV-3DS-CORE-2.2.0] and sends the message to the JaywanDS DS.
7. The JaywanDS DS sends a request for detokenize () to NTS (Token Service), to validate the token data
8. NTS will respond with realPan and expiry date.
9. The JaywanDS DS forwards the AReq by replacing the token and token related data by the PAN and card data to the Issuer ACS as follow.
10. The ACS sends the ARes to the JaywanDS DS with transStatus=C
11. The JaywanDS DS forwards the ARes to the 3DS Server.
12. The 3DS requestor environment triggers the Frictionless flow.
13. The merchant initiates the authorization flow to the Acquirer.
14. The Acquirer sends the authorization message with the authentication value. Among all ISO 8583 fields to Payment Network Switch.
15. The Payment Network Switch requests a detokenization to NTS to retrieve the PAN.
16. NTS will respond with RealPan and ExpiryDate details.
17. The Payment Network switch update the Primary Account Number and Date, Expiry in the ISO 8583 message before to send it to the issuer host.
18. The issuer host creates ISO 8583 response to CB-UAE Switch
19. The Payment Network Switch sends the ISO 8583 response to the acquirer
20. The acquirer system returns the transaction status to the merchant (Token Requestor)

10 Project Implementation for Participants

This section describes the JaywanDS Implementation Project for Participants, which is the process for an Issuer or Acquirer to connect their ACS or 3DSS to JaywanDS and deploy JaywanDS for their cardholders.

The Issuer or Acquirer will have the choice of having their own or a third-party ACS or 3DSS provider.

10.1Prerequisites

The prerequisite must be achieved by the Provider before any JaywanDS 3DS Product Certification Testing activity can begin.

Prerequisite	Description
EMVCo Certification	The Provider which operates the ACS or 3DSS through JaywanDS program must have achieved the EMVCo 3DS compliance and have its product LoA active or up to date.
PCI-DSS Compliance	The Provider which operates the ACS or 3DSS through JaywanDS program must achieved the 3DS PCI compliance and have a proof of the assessment.
PCI-3DSS Compliance	The Provider which operates the ACS through JaywanDS program must achieved the PCI 3DS security requirements.

10.2 JaywanDS OnBoarding Process

This section shows and explains the different phases of the JaywanDS on-boarding process.



Figure 4 JaywanDS On-Boarding Process

10.2.1 Registration

This section describes the steps required to register as a JaywanDS Service Provider.

For the launch of their EMV 3DS solution, Payment put in place a new service to on-board their stakeholders. Each product provider that supports transactions by means of a 3-D Secure Program and who wants to connect and operate its EMV 3DS product in JaywanDS ecosystem must be registered as 3-D Secure service providers in the JaywanDS on-boarding program.

JaywanDS Service Providers are entities that operates as follows:

- Manage a 3DS Server (3DSS) system that facilitates communication (through the EMVCo 3DS specification and messages) to initiate cardholder authentication under the JaywanDS program and rules.
- Manages an Access Control Server (ACS) system that verifies (through the EMVCo 3DS specification and messages) whether authentication is available for a card number and device type and authenticates specific cardholders under the JaywanDS program and rules.

Registering as a service provider is a mandatory step to starting the onboarding process with JaywanDS and its certification process. The registration phase will act as a trigger to kick off the implementation of the JaywanDS 3DS Service.

Steps	Definition
1	The Service Provider will initiate the registration process with JaywanDS.
2	The Service Provider will have to provide all the documents and information requested by JaywanDS for their review: • 3DSS or ACS onboarding form • EMVCo 3DS Letter of Approval (LoA) • PCI 3DS Attestation of Compliance (AOC) • PCI DSS Attestation of Compliance (AOC) • and any other document later required by JaywanDS
3	The JaywanDS team after review will confirm back to Service Provider of completed registration and provide next steps.
4	The Service Provider will receive the following: ▪ Link to get access to JaywanDS testing platform. ▪ Credentials that will be used to log on to JaywanDS testing platform (Example: Company ID, Password...). ▪ EMV 3DS documents such as Specifications, Implementation Guidelines, test platform guidelines, and any other document later required will be provided by JaywanDS team. ▪ Service Provider assigned data by JaywanDS accordingly to the product (Example: 3DS Server Operator ID, ACS Operator ID, 3DS Requestor name, 3DS Requestor ID....).

10.2.2 Certification Process

After shifting the project phase from Registration to Certification, the Provider will be allowed to start testing with JaywanDS test platform. Access will be granted, and information will be shared in a test platform guideline.

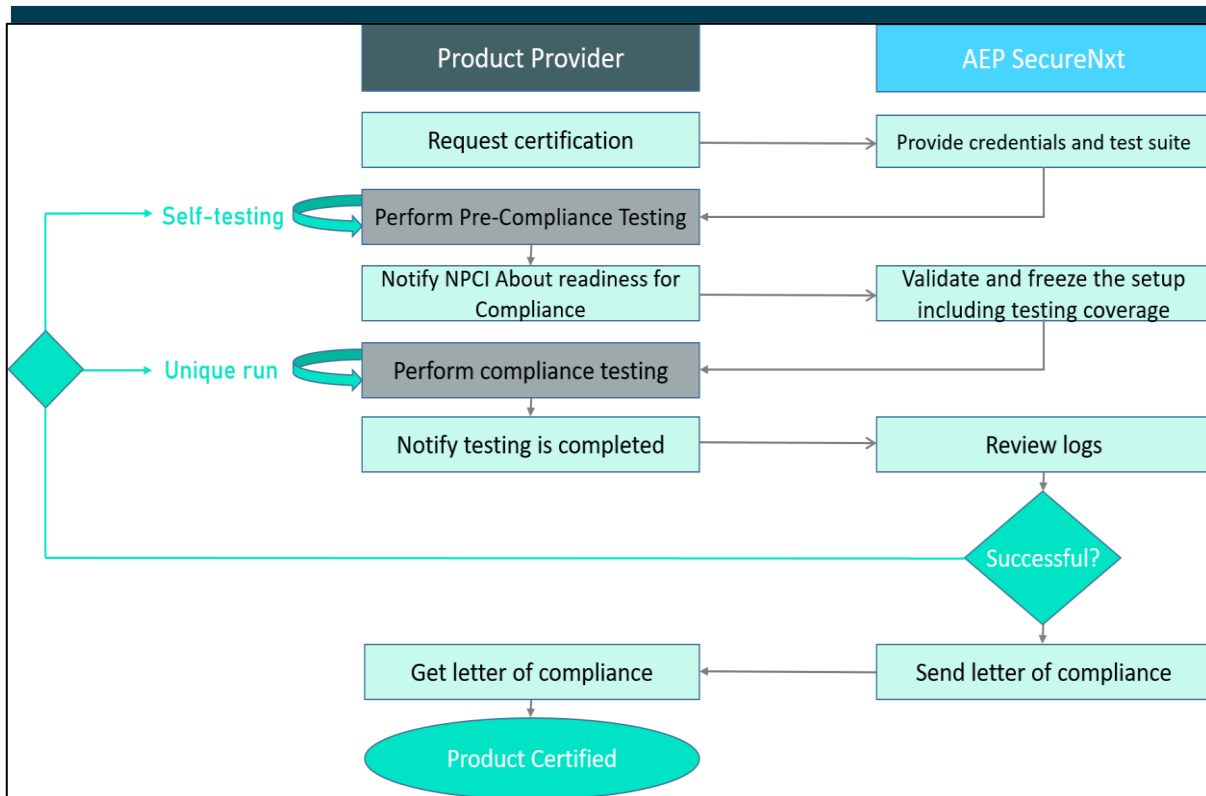


Figure 5 AEP JaywanDS Certification Process

10.2.3 Deployment

After the completion of the JaywanDS certification phase and obtention of the certification letter, the product provider will receive all the instruction to download the JaywanDS Production certificates, then to configure and to connect to the production environment.

1. Download JaywanDS Production Certificates
2. Connection to JaywanDS Production environment

NOTE: Some testing procedure such as authentication transaction to validate the connection and interoperability might be required to be performed by the product provider before the Go-to live phase.

10.3 ACS Onboarding information

This section provides specific information about settings that should be configured/applied during the OnBoarding Process.

10.3.1 ACS Reference Number

The ACS Reference Number is a unique identifier assigned by the EMVCo Secretariat upon Testing and Approval. When an ACS vendor certifies their ACS software with EMVCo, the vendor will receive an ACS Reference Number for the software version certified and, amongst other things, it will specify the EMV 3-D Secure Specifications against which it was certified, the approval date and approval expiration date.

EMVCo maintains a list of approved ACS products on their web site.

Provider are required to provide to JaywanDS the ACS Reference Number for the ACS software that they are intending to enrol and deploy in JaywanDS ecosystem as part of the on-boarding process.

10.3.2 Card range data

The Issuer will have to provide the information about the card ranges participating to JaywanDS program.

- Card ranges participating
- Supported protocol versions
- ACS Information Indicator
- ThreeDSMethodURL

10.3.3 ACS Operator ID

The ACS Operator ID is assigned by the Directory Server and the authority it belongs (Payment Network in the case of JaywanDS program). Each Directory Server can provide a unique ID to each ACS on an individual basis.

10.3.4 Signed Certificate for TLS connection.

The Transport Layer Security (TLS) connection is required between 3DS modules to ensure:

- Server authentication, to ensure that the server target can be trusted.
- Client authentication, for the server to validate that the client requesting the connexion can be trusted.
- The confidentiality and integrity of the data exchanges through an encryption layer.

This section provides information about the TLS handshakes between:

- the DS and ACS – Mutual Authentication
- the 3DS Client and ACS – Server Authentication

Mutual Authentication in 3DS

For Mutual Authentication between ACS and DS:

- the DS CA signs the certificates of all the participants to its network (3DSS and ACS)
- the Keys must be RSA (at least 2048 bits long) or ECC (at least 256 bits long).
- The TLS protocol version supported should be v1.2 or higher.
- The certificates must be used on both Server and Client sides
 - Server side: ACS receiving AReq, DS receiving RReq.
 - Client Side: ACS sending RReq, DS sending AReq.
- The specification defined ciphers are:
 - TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256
 - TLS_ECDHE_ECDSA_WITH_AES_128_GCM_SHA256

[**Note:** Above mentioned cipher suites are EMVCo 3DS specs recommended. Please refer Annexure D, Section D.1.1 for more details.]

- 3DS Server, ACS must support both ciphers specifications
- DS must support at least one ciphers specification
- Additional cipher suites may be supported

Server Authentication in 3DS

For Server Authentication between 3DS Client and ACS:

- Used to authenticate the ACS endpoint for Challenges (CReq/CRes messages).
- Needs to be reached by Cardholders' devices (through SDKs for smartphone app or through Browser).
- Needs to rely on a Commercial CA, as we can't ask cardholders to manually add the certificates for each used ACS (therefore for each issuer).

[**NOTE:** For more details, please refers to [EMV-3DS-CORE-SPECIFICATION](#).]

For AReq/ARes

The DS to ACS link for the AReq/ARes messages are established using a TLS protocol with mutual authentication. The public key certificates of both parties are signed by the DS CA.

Type	Information
Preferred cipher suite	TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256
TLS version	>=1.2
TLS mode	Mutual authentication
Root CA	DS CA
Message type	AReq/ARes
Specification reference	[EMV-3DS-CORE-SPECIFICATION] Section 6.1.3.1

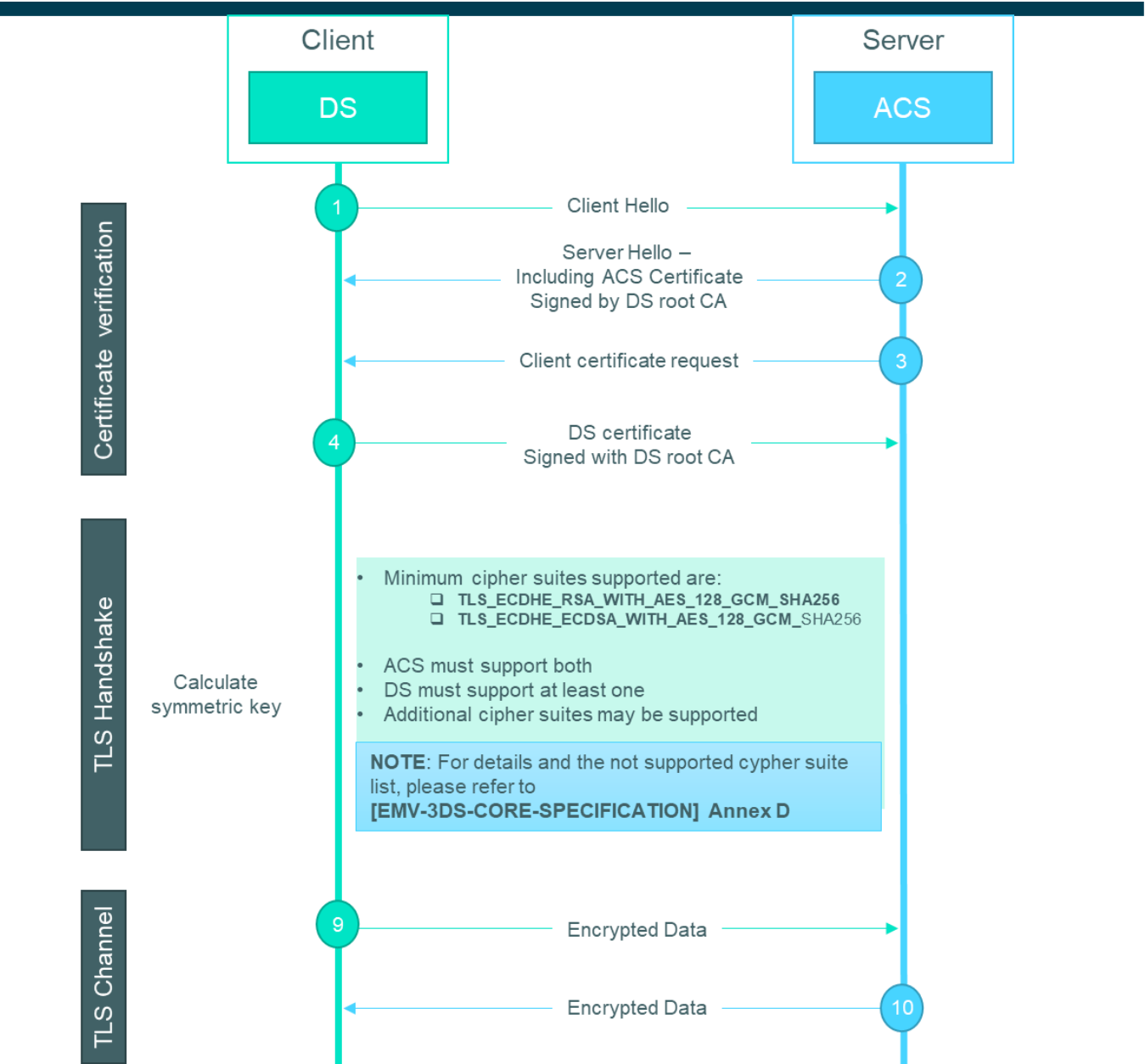
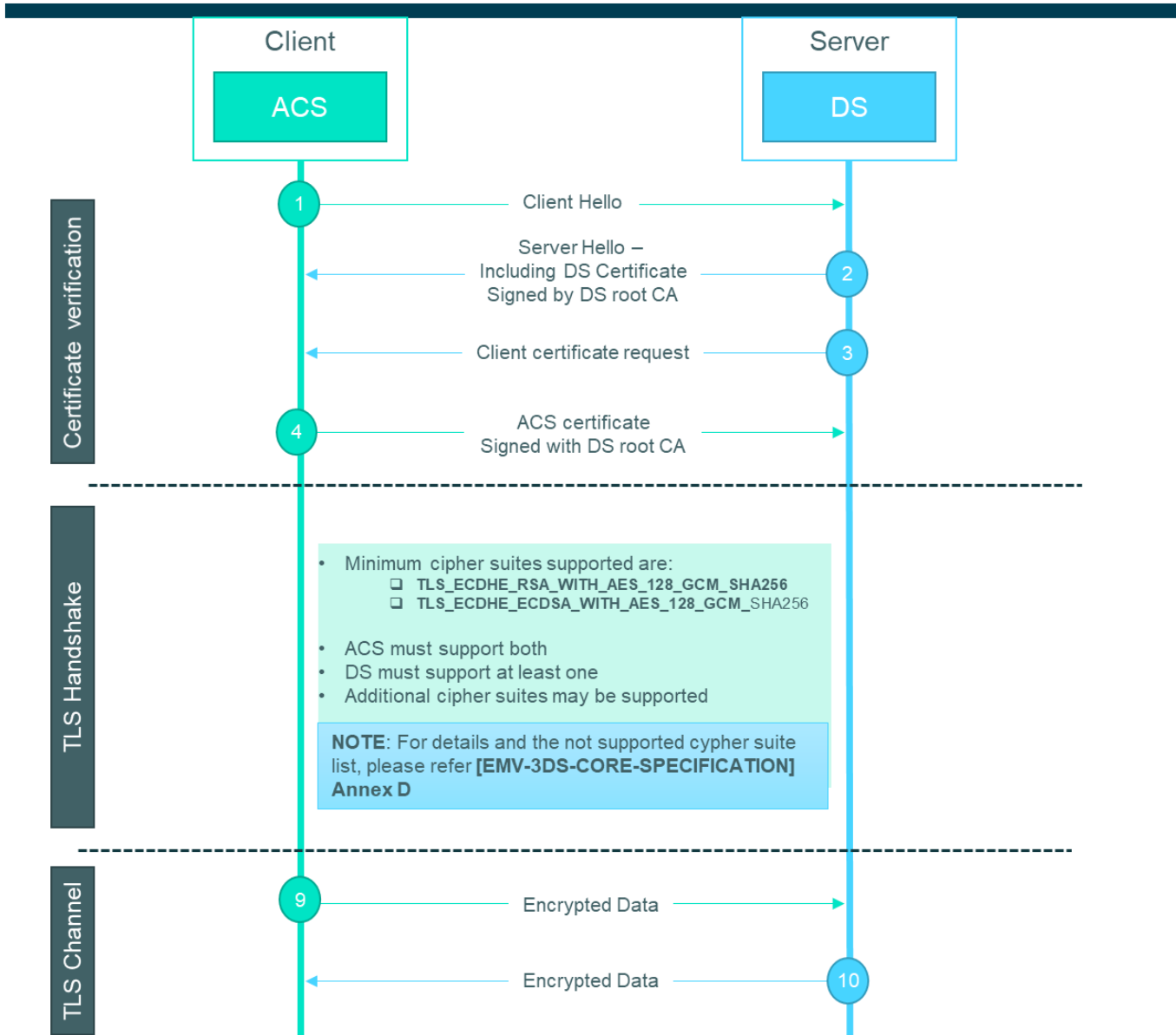


Figure 6 TLS Mutual Authentication DS -> ACS (AReq/ARes messages)

For RReq/RRes

The ACS to DS link for the RReq/RRes messages is established using a TLS protocol with mutual authentication. The public key certificates of both parties are signed by the DS CA.

Type	Information
Provided cipher suite	TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256
TLS version	>=1.2
TLS mode	Mutual authentication
Root CA	DS CA
Message type	RReq/RRes
Specification reference	[EMV-3DS-CORE-SPECIFICATION] Section 6.1.3.2

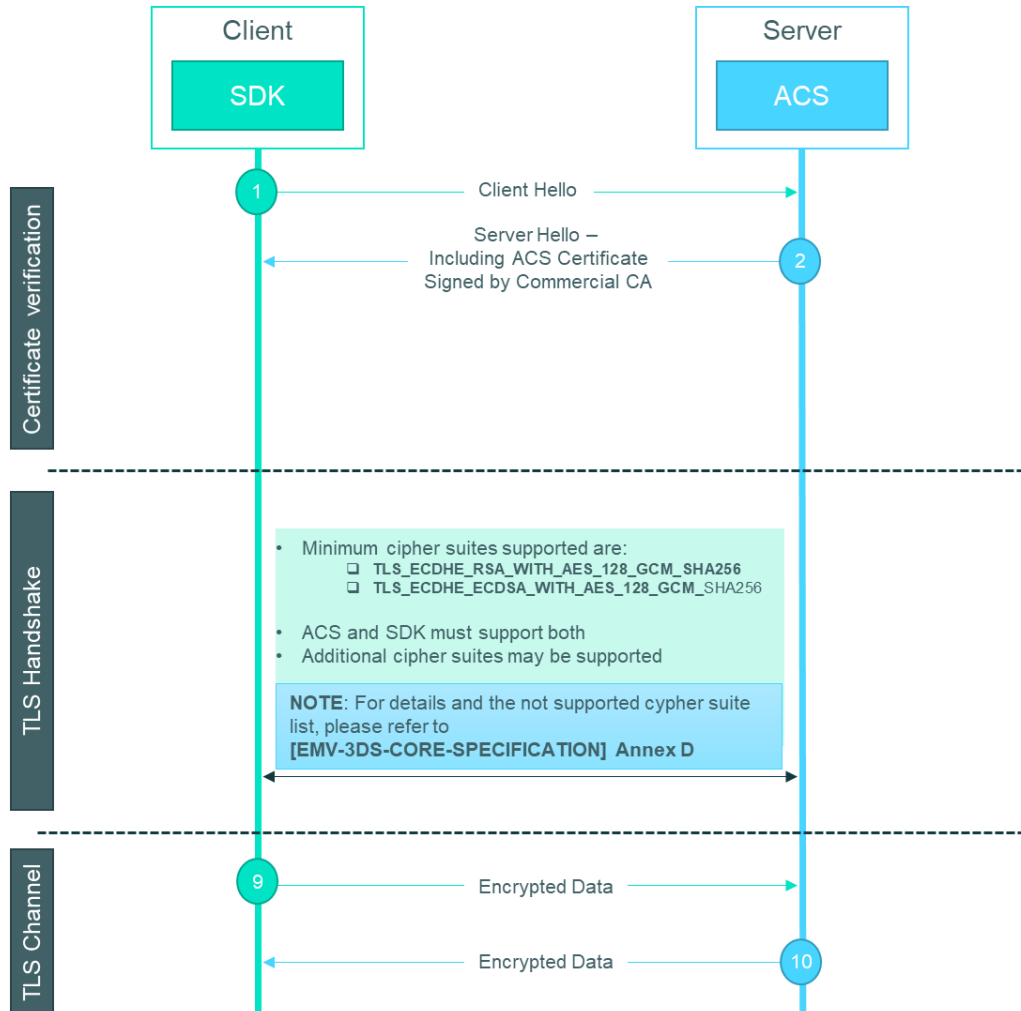


For App-based CReq/CRes

For the App-based protocol, the direct link between the 3DS SDK and the ACS is only established if the transaction requires a challenge. It is initiated by the SDK using the URL provided to it in the ARes and established using a TLS protocol with ACS (server) authentication by the 3DS SDK. The public key certificate for the ACS is signed by a commercial CA.

Type	Information
TLS version	>=1.2
TLS mode	Server authentication
Root CA	Commercial CA

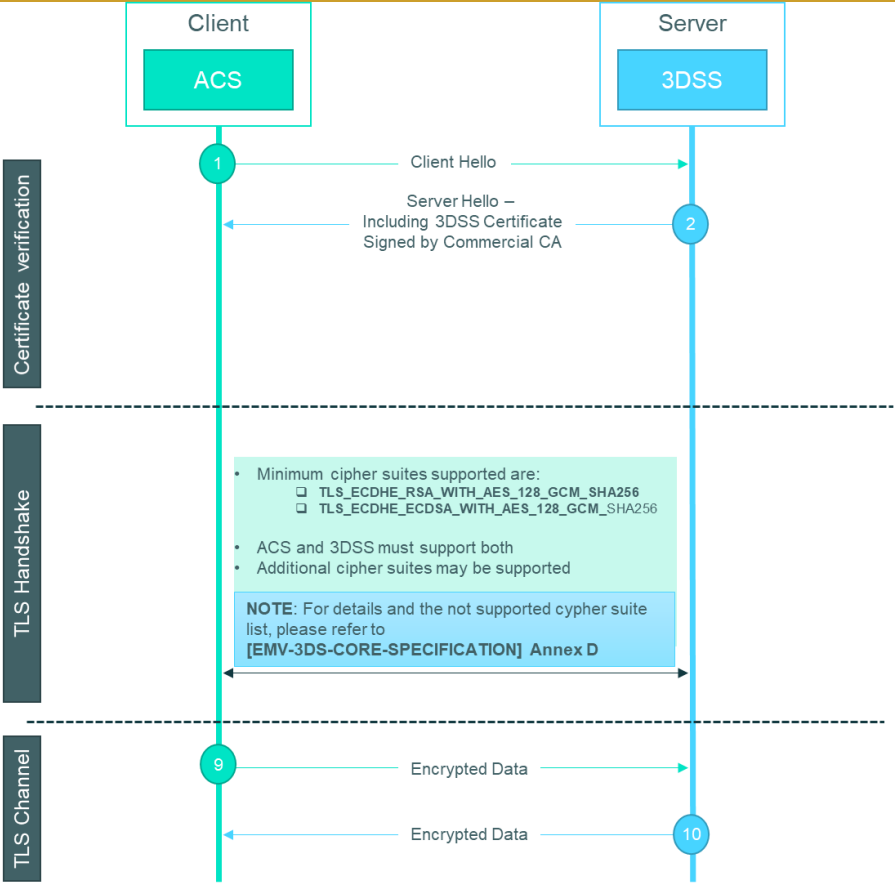
Message type	CReq/CRes
Specification reference	[EMV-3DS-CORE-SPECIFICATION] Section 6.1.4.1



For Browser-based CReq/Cres

For the Browser-based protocol, the direct link between the Browser and the ACS is only established if the transaction requires a challenge. It is initiated by the Browser from an iframe using the URL provided to it in the ARes and established using a TLS protocol with ACS (server) authentication by the Browser. The public key certificate for the ACS is signed by a commercial CA.

Type	Information
TLS version	>= 1.2
TLS mode	Server authentication
Root CA	Commercial CA
Message type	Notification URL
Specification reference	[EMV-3DS-CORE-SPECIFICATION] Section 6.1.4.2



10.3.5 ACS signed content (Certificate for JWS)

For the App-based, in case the ACS determines that a challenge is required, a secure channel between the 3DS SDK and the ACS will have to be established. The setup of this channel will be initiated through the AReq/ARes.

As prerequisites:

- DS public key should be integrated/embedded in the SDK during the built, in order to allow the SDK to verify the chain of trust (DS root CA) during the channel setup.
- ACS has a Key Pair (PvACS, PbACS) signed by DS Root CA
- SDK has generated an EC Ephemeral Key Pair

Type	Information
JOSE	JWS (RFC 7515)
Specification reference	[EMV-3DS-CORE-SPECIFICATION] Section 6.2.3.2

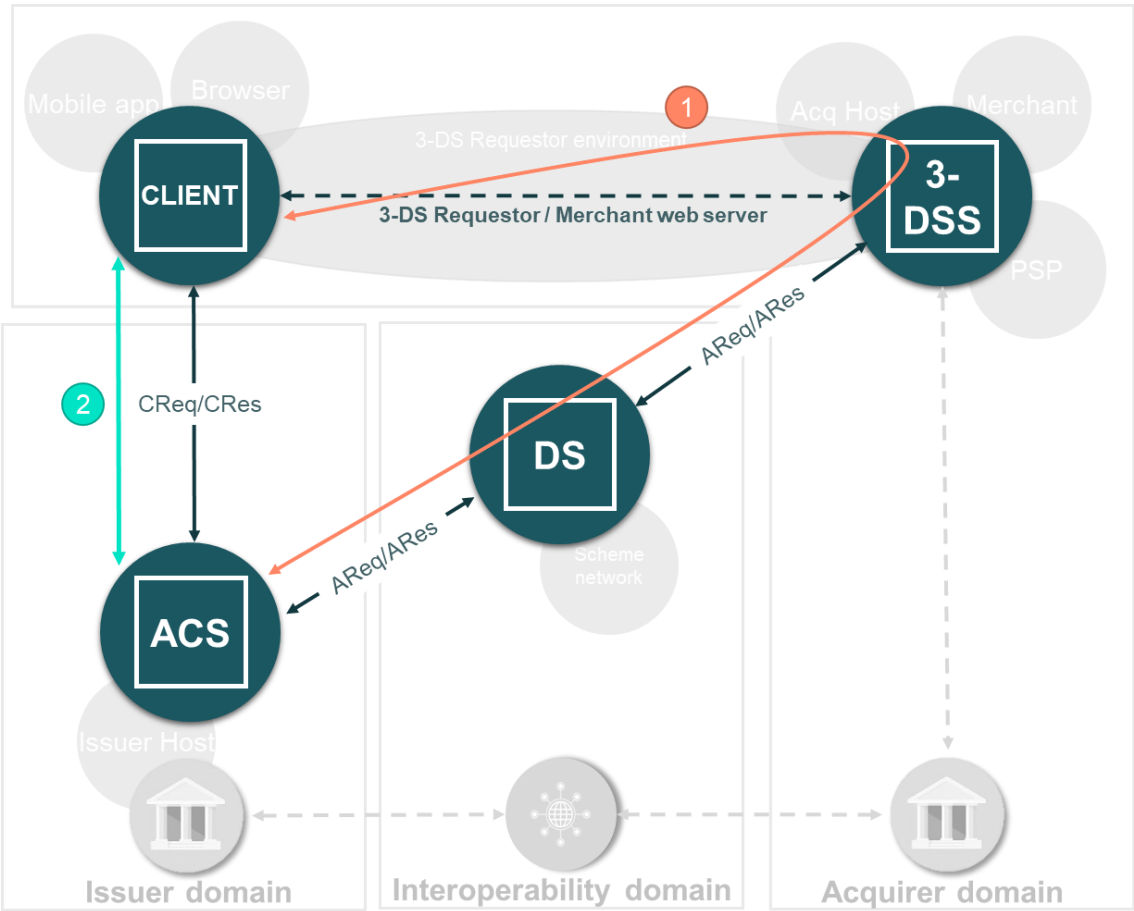


Figure 7 ACS Secure Channel Setup

[1] 3DS SDK—ACS Secure Channel Set-Up - See [\[EMV-3DS-CORE-SPECIFICATION\]](#) Section 6.2.3.

- Key exchange to enable in case of Challenge flow a security channel between ACS and 3DS-SDK (CReq/CRes).
- Keys provided through AReq/ARes messages.
- Diffie Hellman key exchange; JWA (RFC 7518).

[2] 3DS SDK—ACS (CReq, CRes) - See [\[EMV-3DS-CORE-SPECIFICATION\]](#) Section 6.2.4.

- Enables the secure channel for CReq/CRes messages exchange.
 - SDK retrieves acsEphemPubKey from acsSignedContent (retrieved from proprietary channel).
 - SDK does ECDH with ACS Pb, SDK Pv on P-256 curve to obtain a Shared Secret.
 - SDK performs ConcatKDF to generate a 256 bits CEK (Content Encryption key).
 - SDK deletes its Ephemeral Key Pair.
- JSON Web Encryption (JWE - RFC 7516) using the key obtained in Function **[1]**.
 - Two different algorithms are supported (See the table below for details)
 - A128CBC-HS256

A128GCM

A128CBC-HS256	A128GCM
• Full CEK is used to encrypt and decrypt. • Full CEK is used on both ways: ◦ SDK->ACS ◦ ACS->SDK	• CEK is split in half • Encryption parameters are different for SDK->ACS and ACS->SDK. • SDK->ACS:

Initialization Vector is a fresh 128-bit random data	- CEKs-a is leftmost 128 bits of CEK - Initialization Vector is counter for SDK->ACS ('sdkCounterStoA') padded to the left with '00' bytes - ACS->SDK: - CEKa-s is rightmost 128 bits of CEK - Initialization Vector is counter for ACS->SDK ('acsCounterAtoS') padded to the left with 'FF' bytes. NOTE: Same key is still used for encryption and decryption.
--	---

10.43DSS Onboarding information

This section provides specific information about settings that should be configured/applied during the OnBoarding Process of 3DSS.

10.4.1 3DS Server Reference Number

The 3DS Server Reference Number is a unique identifier assigned by the EMVCo Secretariat upon Testing and Approval. When a 3DSS vendor certifies their 3DSS software with EMVCo, the vendor will receive a 3DS Server Reference Number for the software version certified and, amongst other things, it will specify the EMV 3-D Secure Specifications against which it was certified, the approval date and approval expiration date.

EMVCo maintains a list of approved 3DSS products on their web site.

Provider are required to provide to JaywanDS the 3DS Server Reference Number for the 3DSS software that they are intending to enrol and deploy in JaywanDS ecosystem as part of the on-boarding process.

10.4.2 Card Range Data

The Issuer will have to provide the information about the card ranges participating to JaywanDS program. This includes (Refer to [EMV-3DS-CORE-2.2.0] Table A.6 and Req 520 & Req 5211 in the section [8. Annexes]):

- Card ranges participating
- Supported protocol versions
- ACS Information Indicator
- ThreeDSMethodURL
- And any additional element requires by JaywanDS.

The Merchant, Acquirer and 3DSS Provider SHALL be able to interpret the card range data information that will be shared in the PRes message.

10.4.3 3DS Server Operator ID

The 3DS requestor ID is an identifier defined by JaywanDS that is assigned by the 3DS Server Operator to the Merchant at the time of the Merchant On-Boarding.

10.4.4 3DS Requestor Name

The 3DS Requestor Name is identifier defined by JaywanDS and provided to each 3DS Requestor on an individual basis.

10.4.5 Acquirer BIN.

The Acquirer BIN is assigned by JaywanDS and will be unique for each Acquirer and should be included by the 3DSS in the AReq message.

10.4.6 Signed Certificate for TLS connexion

The Transport Layer Security (TLS) connection is required between 3DS modules to ensure:

- Server authentication, to ensure that the server target can be trusted.
- Client authentication, for the server to validate that the client requesting the connexion can be trusted.
- The confidentiality and integrity of the data exchanges through an encryption layer.

This section provides information about the TLS handshakes between:

- the DS and 3DSS – Mutual Authentication

10.4.6.1 Mutual Authentication in 3DS

For Mutual Authentication between 3DSS and DS:

- the DS CA signs the certificates of all the participants to its network (3DSS and ACS)
- the Keys must be RSA (at least 2048 bits long) or ECC (at least 256 bits long).
- The TLS protocol version supported should be v1.2 or higher.
- The certificates must be used on both Server and Client sides
 - Server side: 3DSS receiving RReq, DS receiving AReq/PReq.
 - Client Side: 3DSS sending AReq/PReq, DS sending RReq.
- The specification defined ciphers are:
 - TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256
 - TLS_ECDHE_ECDSA_WITH_AES_128_GCM_SHA256
- 3DS Server, ACS must support both ciphers specifications
- DS must support at least one ciphers specification
- Additional cipher suites may be supported

NOTE: For more details, please refers to [EMV-3DS-CORE-2.2.0].

10.4.6.2 For RReq/RRes

The DS to 3DSS link for the RReq/RRes messages are established using a TLS protocol with mutual authentication. The public key certificates of both parties are signed by the DS CA.

Type	Information
Preferred cipher suite	TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256
TLS version	>=1.2
TLS mode	Mutual authentication
Root CA	DS CA
Message type	RReq/RRes
Specification reference	[EMV-3DS-CORE-2.2.0] Section 6.1.3.1

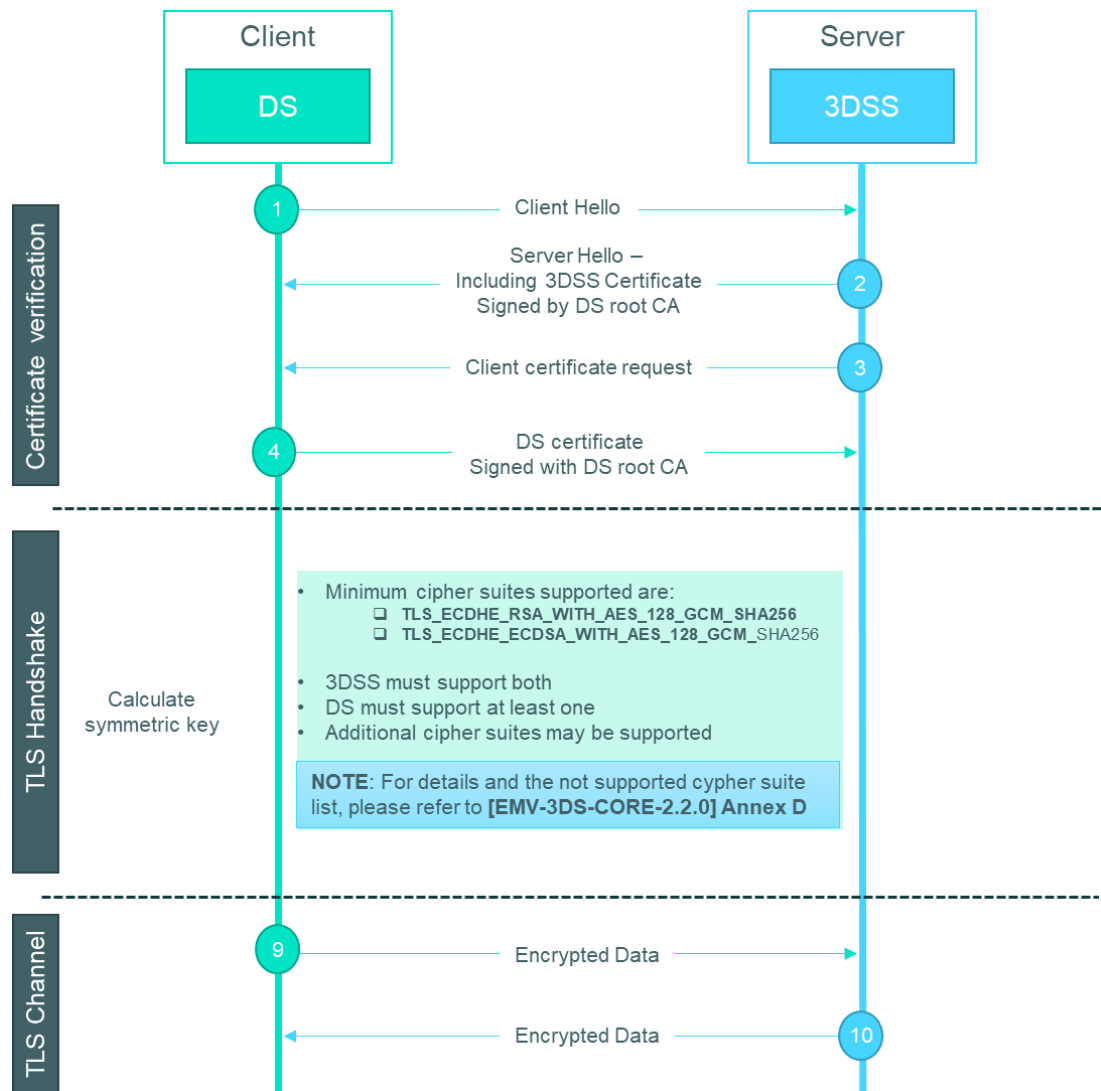
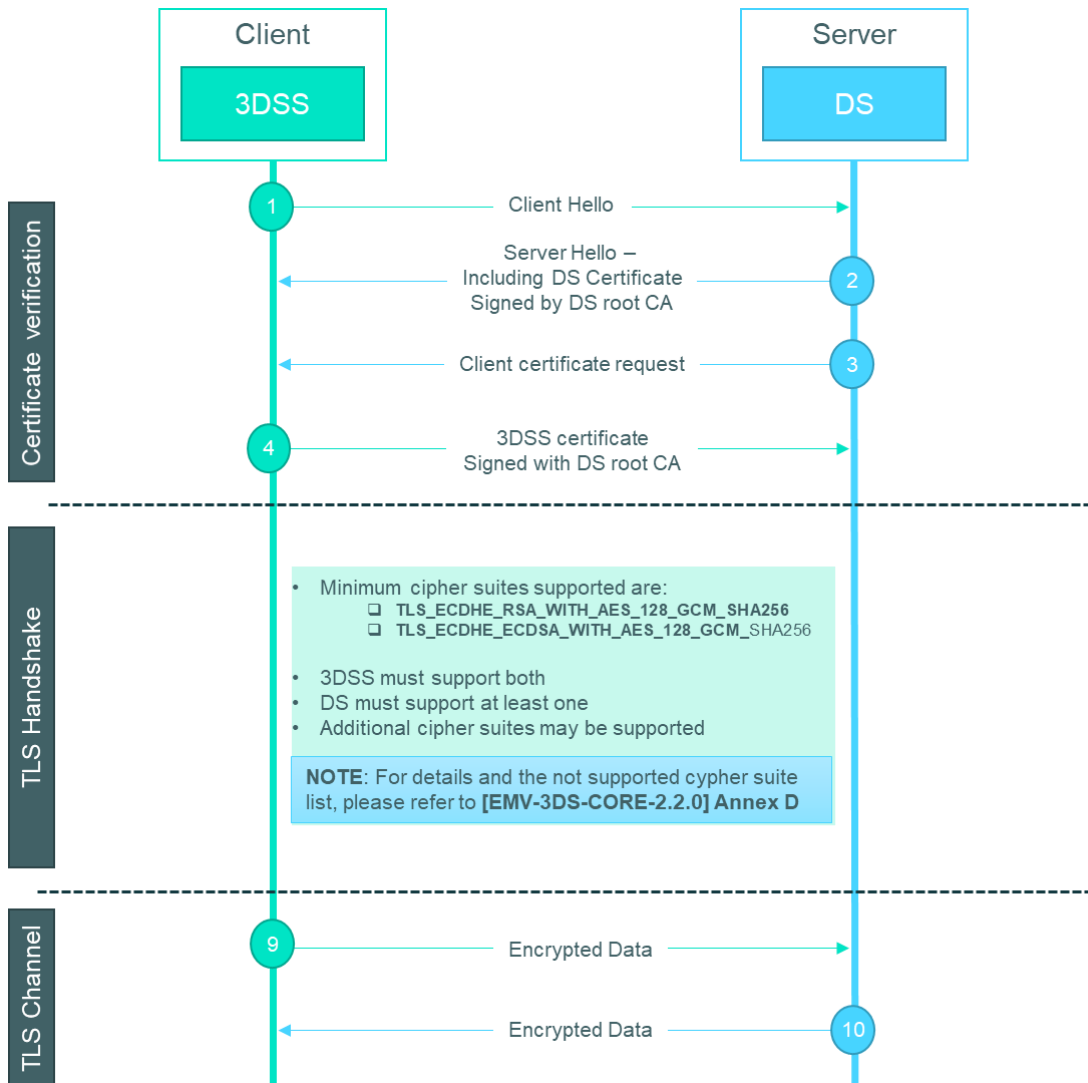


Figure 8 TLS Mutual Authentication DS -> 3DSS (RReq/RRes messages)

10.4.6.3 For AReq/ARes and PReq/Pres

The 3DSS to DS link for the AReq/ARes and PReq/Pres messages is established using a TLS protocol with mutual authentication. The public key certificates of both parties are signed by the DS CA.

Type	Information
Provided cipher suite	TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256
TLS version	>=1.2
TLS mode	Mutual authentication
Root CA	DS CA
Message type	AReq/ARes and PReq/Pres
Specification reference	[EMV-3DS-CORE-2.2.0] Section 6.1.3.2



11 Timeouts

3D Secure eco-system has many components like 3DS Requestor, 3DS Server, Directory Server and ACS communication with each other completing the 3DS Flow.

There are basically two type of timeout that may occur between different components while trying to exchange the messages.

All timeout details in this section shall be read in conjunction with section **[5.5 Timeouts – EMVCo_3DS_Spec_V220]** for detailed information.

11.1 Transaction Timeout

Transaction Timeouts provide the maximum amount of time that a 3-D Secure transaction should be considered valid.

11.2 Connection Timeout

The connection timeout is the timeout in making the initial connection; i.e. completing the TCP connection handshake.

11.3 Read Timeout

It is the number of seconds that the client could wait for the server to send response after connection had been made and request has been send successfully.

11.4 Timeout Values

Txn Type	Call	Flow	
Decouple	RReq	3DS → DS → ACS	Decouple max timeout + 30 Seconds
Transaction Timeout			
ACS waiting for CReq after successful ARes	CReq	ACS → 3DS Requestor	30 Seconds
App Based, Multiple CReq/CRes	CReq	ACS → 3DS SDK	10 Min (600 sec) between each successful CRes and expected subsequent CReq
Browser Based	Challenge	ACS → Browser	10 Min (600 sec) for cardholder to complete the transaction.
Read Timeout			
3DS SDK Challenge	CReq	3DS SDK → ACS	10 Seconds (CRes read timeout)
3DS SDK Complete Challenge Cycle	CReq/CRes	3DS SDK → ACS	Min 5 Minutes (from First CReq to Final CRes from ACS) – Read Timeout *It should cover the entire challenge life cycle so set the value accordingly.
Authentication Result	RReq	ACS → DS	5 Seconds (RRes read timeout)
Authentication Result	RReq	DS → 3DS Server	3 Sec, read timeout (RRes)
3DS Method Call	Browser prior to AReq	Browser → ACS	5 Sec
Connection Timeout			
Authentication Result	RReq	ACS → DS	After first connection fail, immediately try second connection and After second try

			for connection timeout, wait 10 seconds for each subsequent retry and continuously try to send Txn Status = U
--	--	--	---

12 Error Codes

3D Secure flow have different component participating in the flow and processing the messages. There could be different stages and component at which error may occur.

12.1Error Component

3DS SDK	C
3DS Server	S
Directory Server	D
ACS	A

12.2Error Codes

For generic error code, please refer Section **[A.5.5 Error Code, Error Description, and Error Detail, EMVCo_3DS_Spec_V220]** and these error codes can be send by different 3D Secure component as applicable with error component indicating the source which identified the error.

A	03	DS → 3DS Server	Consumer Device not supported by ACS	Not applicable when decoupled

				authentication is used.
A	14	ACS → SDK	Not received CReq after ARes, so RReq send with error code 14	Challenge Cancellation Indicator = 5

13 Sample Messages (3DSS & ACS)

This sections provides the sample request and response message format for different calls in 3DS flow which are processed by Directory Server.

These examples are sample in nature and for detailed information on each data element and their properties, please refer Annexure A (3-D Secure Data Elements) and Annexure B (Message Format).

[Note: This guide and [EMVCo_3DS_Spec_v220_122018](#) SHALL be read in conjunction with each other for detailed information.]

13.1 Message for 3DSS

13.1.1 PREQ Message from 3DSS – DS

```
{
  "threeDSServerRefNumber": "3DS_LOA_SER_PPFU_020100_00008",
  "threeDSServerOperatorID": "1jpeeLAWgGFgS1Ri9tX9",
  "threeDSServerTransID": "8a880dc0-d2d2-4067-bcb1-b08d1690b26e",
  "messageType": "PReq",
  "messageVersion": "2.2.0",
  "serialNum": "66IM7xWlnwjdqG1SSFk5"
}
```

13.1.2 PRES Message from DS – 3DSS

```
{
  "threeDSServerTransID":"8a880dc0-d2d2-4067-bcb1-b08d1690b26e",
  "dsTransID":"f25084f0-5b16-4c0a-ae5d-b24808a95e4b",
  "messageType":"PRes",
  "messageVersion":"2.2.0",
  "serialNum":"3q9oaApFqmznys47ujRg",
  "dsStartProtocolVersion":"2.2.0",
  "dsEndProtocolVersion":"2.2.0",
  "cardRangeData":[{
    "startRange":"10000000000000000",
    "endRange":"10000000000005000",
    "actionInd":"A",
    "acsStartProtocolVersion":"2.2.0",
    "acsEndProtocolVersion":"2.2.0",
    "threeDSMethodURL":"https://www.acs.com/script"
  }, {
    "startRange":"20000000000000000",
    "endRange":"20000000000004000",
    "dsStartProtocolVersion":"2.2.0",
    "dsEndProtocolVersion":"2.2.0",
    "actionInd":"D",
    "acsStartProtocolVersion":"2.2.0",
    "acsEndProtocolVersion":"2.2.0",
    "threeDSMethodURL":"https://www.acs2.com/method"
  }]
}
```

13.1.3 AREQ Message from 3DSS – DS (Browser Flow)

```
{
  "acquirerBIN": "000361603",
```

```

"acquirerMerchantID": "1000",
"addrMatch": "Y",
"billAddrCity": "Oklahoma City",
"billAddrCountry": "840",
"billAddrLine1": "54 E Pear St",
"billAddrPostCode": "73102",
"billAddrState": "OK",
"browserAcceptHeader":
"text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;
q=0.8,application/signed-exchange;v=b3;q=0.7",
"browserIP": "174.91.51.140",
"browserJavaEnabled": false,
"browserJavascriptEnabled": true,
"browserLanguage": "en-PH",
"browserColorDepth": "24",
"browserScreenHeight": "960",
"browserScreenWidth": "1536",
"browserTZ": "300",
"browserUserAgent": "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/537.36
(KHTML, like Gecko) Chrome/120.0.0.0 Safari/537.36",
"cardExpiryDate": "2608",
"cardholderName": "John Smith",
"deviceChannel": "02",
"mcc": "5942",
"merchantCountryCode": "246",
"merchantName": "CoffeeHouse - Discover-3",
"messageCategory": "01",
"messageType": "AReq",
"messageVersion": "2.2.0",
"notificationURL": "https://3ds2-mpi-
discover.test.modirum.com/mdpaympi/MerchantServer/msgid/34610411",
"purchaseAmount": "10000",
"purchaseCurrency": "840",

```

```

"purchaseExponent": "2",
"purchaseDate": "20240131110445",
"shipAddrCity": "Oklahoma City",
"shipAddrCountry": "840",
"shipAddrLine1": "54 E Pear St",
"shipAddrPostCode": "73102",
"shipAddrState": "OK",
"threeDSComplnd": "N",
"threeDSRequestorAuthenticationInd": "01",
"threeDSRequestorID": "1000",
"threeDSRequestorName": "CoffeeHouse - Discover-3",
"threeDSRequestorURL": "https://www.modirum.com",
"threeDSServerRefNumber": "3DS_LOA_SER_MOMD_020301_00816",
"threeDSServerOperatorID": "DFSOperator",
"threeDSServerTransID": "9c220122-fb10-50cb-8000-000002101ceb",
"threeDSServerURL": "https://discover-dshub.test.modirum.com:9807/dshub/DServer",
"acctNumber": "****"
}

```

13.1.4 ARES Message from DS - 3DSS (Browser Flow)

```

{
  "acsTransID": "efa74225-be59-4a91-8553-08aa275774b8",
  "acsChallengeMandated": "Y",
  "acsOperatorID": "acsOperator",
  "acsReferenceNumber": "3DS_LOA_ACS_FTPT_FIME14_0009",
  "authenticationType": "01",
  "dsReferenceNumber": "3DS_LOA_DIS_FTPT_FIME14_1101",
  "dsTransID": "4b6a43f9-61c4-4e42-a230-d80b5c829776",
  "eci": "05",
  "messageType": "ARes",
  "messageVersion": "2.2.0",

```

```

"threeDSServerTransID": "9c220122-fb10-50cb-8000-000002101ceb",
"transStatus": "C",
"acsURL": "https://qccert.aep.org.in/ds/sandboxdssim/acsCReqBrowserAuthentication"
}

```

13.1.5 RREQ Message from DS – 3DSS (Browser Flow)

```

{
  "acsTransID": "efa74225-be59-4a91-8553-08aa275774b8",
  "authenticationType": "02",
  "authenticationValue": "QXV0aGVudGljYXRpb24gdmFsdWU=",
  "dsTransID": "4b6a43f9-61c4-4e42-a230-d80b5c829776",
  "interactionCounter": "01",
  "messageCategory": "01",
  "messageType": "RReq",
  "messageVersion": "2.2.0",
  "threeDSServerTransID": "9c220122-fb10-50cb-8000-000002101ceb",
  "transStatus": "Y"
}

```

13.1.6 RRES Message from 3DSS – DS (Browser Flow)

```

{
  "acsTransID": "efa74225-be59-4a91-8553-08aa275774b8",
  "dsTransID": "4b6a43f9-61c4-4e42-a230-d80b5c829776",
  "messageType": "RRes",
  "messageVersion": "2.2.0",
  "resultsStatus": "01",
  "threeDSServerTransID": "9c220122-fb10-50cb-8000-000002101ceb",
  "threeDSServerOperatorID": "DFSOperator"
}

```

```
}
```

13.1.7 AREQ Message from 3DSS - DS (APP Flow)

```
{
  "acquirerBIN": "000361603",
  "acquirerMerchantID": "1000",
  "cardExpiryDate": "2608",
  "deviceChannel": "01",
  "deviceRenderOptions": {
    "sdkInterface": "03",
    "sdkUiType": [
      "01",
      "02",
      "03",
      "04",
      "05"
    ]
  },
  "mcc": "5942",
  "merchantCountryCode": "246",
  "merchantName": "CoffeeHouse - Discover",
  "messageCategory": "01",
  "messageType": "AReq",
  "messageVersion": "2.2.0",
  "purchaseAmount": "13100",
  "purchaseCurrency": "840",
  "purchaseExponent": "2",
  "purchaseDate": "20240126020726",
  "sdkAppID": "2d5a1974-f87c-4ad4-81f8-ebd558772133",
  "sdkEphemPubKey": {
    "kty": "EC",
    "crv": "P-256",
```

```

"x": "sUnQLe2WuAOJWCsMVzOJ_HMfjeUEG087Y_nx9I9ESNk",
"y": "UPLd24K35B1RtJvYkzSedLEKWPNvJUxLosebFelqyt8"
},
"sdkMaxTimeout": "05",
"sdkReferenceNumber": "3DSSDK74332823FF",
"sdkTransID": "c8dfcdf2-2c6a-415d-b520-4506f7f3b9eb",
"threeDSRequestorAuthenticationInd": "01",
"threeDSRequestorChallengeInd": "03",
"threeDSRequestorID": "1000",
"threeDSRequestorName": "CoffeeHouse - Discover",
"threeDSRequestorURL": "https://www.modirum.com",
"threeDSServerRefNumber": "3DS_LOA_SER_MOMD_020301_00816",
"threeDSServerOperatorID": "DFSOperator",
"threeDSServerTransID": "fa76b3c7-e3e9-54eb-8000-0000020fd9cf",
"threeDSServerURL": "https://discover-dshub.test.modirum.com:9807/dshub/DServer",
"sdkEncData":
"eyJlbmMiOiJBMTI4R0NNliwiYWxnIjoieUINBLU9BRVAtMjU2Ln0.PqkMMKaVsHUcyOFiT3OYT5iC87
ANzKISPyDTL_D_jRfDUle-ykwu6kpDXoK9AM3-pgj8fVM_g2kdzNuGRzqRIgqtA0jwEKMM4eUKt-
mkO4P_RbXzd62CdW8bV1rgpEia34OpiumTykj3LDoKgyba1XiKxQoVVT0Axx5-
LX2dO1WaePVY18myuAuaH-Ob-
zURipXmDjl4Q4zHUY7zVE0Ccltj3JXAGX9v6skRXNly4ftMu67cLN1HKWcht-
8X5NWFD7iHHztT_ewGaGDWz0WQsbB9QkYJP2zIS_CBXBSDqb2_OXmkJ12SnriehzdNun_r2-
4P-HA0THbuk0WMYirXPQ.p-
wvTgU3nGj2UKxO.kH5WTH0Qq20iSfJYHJv11BGPJbq4GxQupXlzfIRFNj7VH3GGaOnjeEBvnEW9a
DiNYIVOy-
o10pdMHLTVaUrLtHzH7_KyumKrLxlrAmczRXpdxjzH4_42TcyLxqMHdh8yTWmu7G8Y2byO1zjaqf
Q_IQDIzBiFgs6am5XtwCQvD7ON6Unhus0BTdGka4kKmP21BBmnlZcbbk-WCOYFqD26jhNX-
GISuXLON0KOvaxEtj5Mmg89FZyq5pAlCMnXldnKxOveMg-D_EzIlbKGArLws5uVsaP-DT-
pMukLVIYQJHDLr9tLt_SY_OVDN_4G64PtIl_YOowCU1CcP5Kc0HkNmndSMcEN1IKCFDZpaRqxJ5
UzNcFc0yl2g-PuISSrZnAkzna2RxTuyQ90DsSBWYyFvGRVB4Q8wC-rJh2CStoFTuRjYQfp6-
T2s4tCArnii67XsTxovn8brR-oBTjvAiVj0AzyWZzS1KyZj14DilhxqswQtEi2Du9c550pgglwYsV-
9p44xrD_kJYUKY_Yw_fUbeHOrvTpeHX3HOq_yGeimu93k2hyjkiS74QrLP3ugL7clltSU31BZmz84
kSjsOrTeVKHL1kvR7yLzoM9oGrVzIF2-

```

mwLgLdT4KDHfQevw6Z5rXbgCu9X69pMkxMuqXi9oy8RuiwpavlfXWcsdiulEoKWwOz-
iAxpNyCriqaO5PYgdx23R6HB9_Ne_IJdic5T-DwpPbfBOQO-
UciTvW5cbEx6ybRplO6b3Nlx9HPcRHuC2iWz8FnL1gyaDV4aC3hZQzuRTRoF_WqQILhJcTMjSwac
uycWhQ2laR9NDIZdYXsRQy_zkOooLVMZVFg7s1nL9xdzTzNyC5UCeIUw_Ozb5NdTURxIUsmYqu
6gikDxIjSc4_SaHy7XVp31A8VBf10uWEJe88willBj01kwcAAmtE5ra3s2enY2aM4P7WzmzF7EttH
zZSnXHN8jz9h08kZFmMmk_9oNN4Fi0sZdAgb4_XXZJDz-
5nU7AuR4FjGJI55YQ0GA8r2h9pmZpwE_ZQUUU0m8725RHcKjs8MIO1cxSkHu3hRnXMtb4t3Eu7
i5rKu4egmytdTja7yJ-h1uY2qYK1peWTIU5ZxGOHf_and-adeFjIA-
No7Si6lsxnnzz7PWAeM3dSmUCjfcQHWBnikPjjoHEalNMPv33k29fqhK8MmesladcF9fCobnQGUO
bUSig4HX5JGeAOPQImLMIBGpRBVX8M3ICavgL1y2IGUNAM7RJXnyEAe-
1pml4altvjFbsCKMIGjmRXKS8CmGo0S6AUPL6YzJnltChg73clkcqHr2Kn-
t8Ja30wgt1ukGVfUzHvngezhWPmMhCG4xLXqZhTDb63ftKGQ_Oxr2XPT7dBO4wHo0JbHDgAz8u
wWmmCowgzs4qMMVgFJgjb5CRNuBjRT0tTQFrQcUCFaUMBbu6mkkjJZhzwvSyZxWSZISAq8ZJu4
anjKJ2Ubn-Oi4Hjff-
iV6Vv_vtXNe3JsIXG5i9Vwa1y6pFxmJjU3yxxmMYehxyPrDeabZgaiG9_YoHfJBWi2hkcyLhKApOYA
dzVvaptqj-L9TSPrrrC-EfGIBU9FErTZsDYLRhutRQ5t3u3hyIf1P-IX4-
iZldKOMsz24YXFMcMBLfQo4jZ7_EyGcULAT0vdxduBiGRo4ptX-
eVsjAhVRqCDsoQd04pKCvzcX_lokFExy4AADy4xJtgZ-
CX_fll1ukZj6l5Tw37mLCpj63bTCxV7g5ualC_TRV-
NOMXV8Mgwgngnu0AG3Qu_OfHAcEXvXEGe3UHMkSK1o46OD-
za8x3PRAowVLbv8GX7S8Bl4xewpIYnvLLPXxQEwqSja2aAOVS1MN4b0n-
S3dHjWIZlcUIDMAbOyfdAD8dGlbq5_ksZdo3z0DZXr4-
KEYOgolUsG7jroNFnXLIBvbVTLzQORUwH38Pydk9CyS-2svKEz9BW5qg3-
OrFqYEylnUTJxc_oV_qMwSnFF1GDWY7RFTubH4T2twyHfVMImj_QJw5l3f3l4-
MJMfhjql_GAu4vObI83B3m0eWRmIIAUwz5Sk9Pj0dkNBH-i026aVbx1QuKq3dL9sN76orrx-
hf2aqqaqUp-G1Ee-01yJh_Za4Eyf5sLZ5rtxIDsqlQ-
buQl2hidWILa3l1UXvqL7cXeGq7724z3jImdZ48ijmABTmKXUy-
G5ziG5V44NwR7AkZDZ5Ew5_QtVeILB4lp_sOg2VHLriFI1Sk940QFq-
zlgI5YvPJox4GAXJj9G9rvv4LQgUlVub6O0f8L9cKo7QZPzJPRPWmJpLcG6B4c21HPpf7GpD2T0x2w
6TiD1Xmu5plwxYr7q2_f82IroMbWG8kRDjlyqRNJkMbla3voaoJ-
BJGV6bPOX9oLs75HHC8rt1GzgB8zToUBuWu7ynXCGWtc8JJxtA0Q7zGhH9OE5EsftUF4cL_3kkZ2
iS9n8MD7cNag0HwZO4v4JvBfO6Tk9p1jKdu2J_wocRRhoVVTjOGj_TIXGprq05OhCg-h75f-
bHvbic9F5ob71fBIESfCImlJYhYB_EUAPaW8009mUZFU_wcaxprG3lrlKI9_2wZdUVQTsMO1xBv

MPHfV_065HxNU0AmQjbl--9BU7t8ASoYApR6G1bopKWYTjUT6jSCvljzl3qNu-HT81FMt7-
 hyl4ry2gC9Mm96Ue1E44-
 ETDmqS_YtxfpoWkGQrI5NGod8TYf04TqMXGbe1D8lu1jF_TwgU2swwPituESiBRv_v9G9pZPBRs
 sYSXoWiv1j3W_9nvMou63SafhHO_AcukUfoVfmlte60IJ_YYmjnBva7qtA8JqdF4ZRtUkRpMEA0Fg
 EzBv90Yh9ReWs-
 YkPINxt2JDDMmnKY8Dd79zFmzROWqqbaSGE81ciU6dOJdJtUGk4hau5vU2q22-
 GA2gVYv7XnD7uvcZOyLWOMAKr-
 2R7cNGN1yAjU882Ucwvca5jpN9LLA_MFU4CxXdSWPxSRHhXarJOICUkutB59nhZEwwLG_ZjzgAX
 wxhwbCEEyE-
 eFbc5n8y7nnpb7uff06YfYEowtR5FSila8KgLUOAi9R3jiN3oDmoldD3jydRMe_j0VOHj0JSCbB8Mjjb
 AW6mTe0Jd1uPzIHZDM3_5b1vza-
 maXKxztuRaDWBt3UMDrDjlgMxdo0Wif7yHgQK5ulen9WYQd7tHhmTpOcNuAKFYy9xmXGYOQiD
 GUvaOs52z09LrkN7ot9wA4c5gNrBfo1X-
 WPM_4LFmLtbFJTksNyakgZxu7WgjUM64WqR50aNj49L-
 WbPbzbDxT0hxpjlww6MU8oNx4iyRwKqRNXuLDAp-
 zju0IX0BeyY4kt2tYAvBkZPzIRism9m0OPEbCKHpYiCL967DHAjU_I-
 QjxIYW_NcV9bzHyKm_tG9KBYXLkv0iYwaV8n1CLYeRXDOROyK_-
 uloza6IGWhk_V_DkzhMR_Ex-io1gHvwtGeWHyomAMxGSCsQeLp2D-
 eRHpClkJ8Jwb4AatJFSrqdmWVdf6ziLO_9KvhCatyVyNaICH0gigRzDTz7GxhH-HYufCpKr1-
 55I3Q1DJOkR3S7osh9umBZNULmnGhdq7_V24YGxXACyET-VvXinNGYePMj3Gur-
 bGjvS0iRIWY9g-TLnd8OWtnNwMFXJQH0D8L9vNFnwo90_s02-XapRiBiP6vccgm7v1MD-
 k3SApeMmi2oMfQH0c1Sm1eK6DTLefcDyHFF2guTIODVbwez_jl_-
 z1F67LwkBJ5U8tVKeB74ZjATsn62o5vYvmRL0DjmEP9do5ej5E60RODm5vHVOOVySgrVsPd4x5i
 b4ONTBiqN45nB2mT_o0ImAS8Cn68CZbmjv2Y7eckLvKArRyqmSeD0ggQtIYF6I9isgSGgcjloz7HVI
 WAJlp3FdaTS6-
 4OI7z1B41ydvM7IL5iLZNw7cuXUSFoPbTLDppL_PSNcKlgMMav82AQWqYFnzL1przLWKfTR2JwX
 VU58GOMdzdwK_SzvPivRN4VOqPTNAkv8asBSkgaGI8Q2QgUjd8F9cDSGckCLZyBk5pHEMIAag3
 R7Vq8FpQ2qrJOlqbKdds2jPHBYsI4WW8CojRHG36ZzBRgPf7tLdktW5t8WnNI00s2rU00wiEse7p
 B-HUrRqmQxBGixGOFHAK8wFiloiGHwQ30TcSFbl4IsW3Ah1-v5KW5INn7TO5JeFFRrIpSgdC---
 oOnh-
 noSYC9uUtRx6C39NgKX0ex7dsUSlbfBcTeSGj1iael36u1KGC3LIYRNbBpilF3AwHogWZogd1x-
 To5VGcS8ZAlOk_tLdgUb6_6n9_qjcN1HwP3HuhSYGf3qwZoz3pLy9mMdb3qleSkxWfNNtQNv2ct
 53HNqfWOTHdM1v39_me4msRgmM8nyG1a4fZgan3_a-

```
Hq4lGmbHxMMLsC2nGuStyk6COxZ9VGA85EqyYQmGR0ISnhW2FipVkubocJ1y-
NIFMQG1kLoYkq1HTknUr0Jzb5apB3YHgZ7MPreFCnyS3UtUa2yMhywCjfrfR5PC79Ga3IN2FTDEr
D0HpSQlFAk93QQowGWpsnxEO2Us_7VAD-
qtHbeG9v342Dfui8M_I4hd7Z7RaHJ7LFVHNH8sG5buDzg0VggaliYzZuta8M59G3Ers6l9tnfm7vOc
g1yQC3XRmnKojmJUysjqKdgYwR5sWycQufX2Di5N8kYG0FXu_uyB0WQYp7Cef71_vZk_S3XtNz
MCKOU8jUsJ4j2TuNfYrKlcaqlyDuWYQ3KXTAAZQ5DqX3mvfhDauFv1AGBJPa4j8Ez-
Oa2Smq76ES32xtIA99R-GqX8ImpN5D6-U3zW_s9-
DIUperbHcRFUbfUM4JfqC6ZWwFDByskZJmv8IzApLjAT8dPTaLSnMj3yi8ZGbZfwZvXgNXX9lbboue
u7PgEm4mo7vcrm5SLhQqajXZ3Dlu85_sfZ7mNGZMSkgXIVnvFRFQeN9K8VpbVpjhFjMhjBurBMj
bQLnNdZb4yoza1zvt90l75wT2lgKi0RnNsDvcRanF7Lo_70izlyfv365CrTpTO3aLYhsaKtLQSAp85S
OMMiuFmMYNaXFkb21bhPEU8Sfl-lsqal9jBnOhhulT-flMUP1xK_87Ou54zd3RK-
gNgVQFJD1coXVqczk2QSYyRMgZlOQ5WPT9t6rXOBc1Hzj3gPk5JAcgCHnl8QidzYCnwIPYFzHXP3k
99Qo66AYZdFwBDAb0.NMTXxvxSb0xj2B9EN1B9qQ",
  "acctNumber": "****"
}
```

13.1.8 ARES Message from DS -3DSS (APP Flow)

```
{
  "acsTransID": "823012ac-f144-4301-b5f7-f81ef4d412a5",
  "acsChallengeMandated": "Y",
  "acsOperatorID": "acsOperator",
  "acsReferenceNumber": "3DS_LOA_ACS_FTPT_FIME14_0009",
  "acsRenderingType": {
    "acsInterface": "02",
    "acsUiTemplate": "01"
  },
  "acsSignedContent":
  "eyJ4NWMiOiIsiUIJRgd6Q0NBbXVnQXJkFnSUUpBSUFGZ1ZUK2IWMmtNQTbHQ1NxrR1NjYjNEU
  UVCQ3dVQU1GZ3hDekFKQmdOVkjbWVRBa2xPTVFzd0NRWURWUVFJREFKVVRRqRVFNQTRHQT
  FVRUJ3d0hRMGhGVGs1QINURU5NQXNHQTFVRUNd0VUbeJEU1RFTE1BaOdBMVVFQ3d3Q1N
  WUXhEakFNQmdOVkjbTU1CVVJUSUVOQk1CNFhEVEl6TURreE1UQTJNRGswTVZvWERUSTBNRG
```

t4TURBMk1EazBNVm93VORFTE1BaOdBMVVFQmhNQ1NVNHhDekFKQmdOVkJBZ01BbFJPTVJBd
ORnWURWUVFIREfKRFNFV9Ua0ZKTVEwd0N3WURWUVFLREFST1VFTkpNUXN3Q1FZRFZRUU
xEQUpKVkRFT01BdOdBMVVFQXd3RIJGTWdRMEV3Z2dFaU1BMEdDU3FHU0liM0RRRUJBUVVBQ
TRJQkR3QXdnZ0VLQW9JQkFRQ2s1M2tFYVQ2U29SbmdNbFkxdUNrMDI0dFVKR053QUpyakhCO
HIYVXIpaDhwRklLWDYxNjJ0cUVMTEtpRThJTTNjeG9qZ2lLdHc1d0hFQjAzSWJHaEJBWldXdmFwbl
pvQzExSHBveVJOV0Rxb0U3ZGpaSkRzU1N5MXInWW9KY0U1V3Z4a0h4TFpRS05LSXk2RTk1MjQ
1RIJrbDJuTTJPOVhIOWxOV0E4dUJ6NGpZSW1FNIQwRVRzRDlpanBEa1ZiZH4cXBmd3FoaEVJSkd
rVUdadml3dlFEclhJvZcybTZkTjhHZURRMEVZQVZOOU5ES2t4bzZiNzBBY1BZbUpRV2l1N3JXaUz
QWFTN3hpOEhxTWVxMDExczZ6SFIXamhXclgvRE1EZ1VucC9VL1JxbGpLZENJK3JJeW9sM1BOYW
xkQU8yb2dPR2ZkbEM5Qzh0Mjd5RmlSQWdNqkFBR2pVREJPTUlwROExVWREZ1FXQkUdTh1R
WNxNndPR0tLRFVXQzN0dmNPUkNDdC96QWZCZ05WSFNNUdEQVdnQIR1OHVfY3E2d09HS0t
EVVdDM3R2Y09SQ0NOL3pBTUJnTIZIUk1FQIRBREFRSC9NQTBHQ1NxR1NJYjNEUUVcQ3dVQUEO
SUJBUUFsWHd2bktVUnNMUGJIUktjWjBKRmY3c3N5cHJDcVlzdS84dmkzY0d4dzZpU0QwUjdVdTA
OVkthZXYwTU85a0hzd3pnU2JkOTBkaXRYNWJReXpqMjVWeFZydnQ0dk1yR0tJUER3RGNWZDdv
Q0tNNDI2bDY2L2FHSHRUVGQ2QmFReU4venpUbzNPcWY3RmVTdmNDSkdVcUJLb0l1TGtRQjV6
MUhZeG5jU2VpS25lOVBoRU9MU0Y3OWF0S091ZzYwd2NmM3pKbVdleG90eVBMWGVcWHVzb
y94ZEMrNGNUYWZtQ0JRTIFVRUKxU0cwR1czOFpCU2U0ZHRlK1RaSzZGMnBiRW5CZDZ3SE5pSE
dLbXILQ2txT3FyQ01yT3p5R1hiUFB0THZ3OGtxOXJSdXQvM25rMUICNkp2dUFleFFERTZtM2hyU
WxIYjNicDY0eVRJUIFDUGFQVYjdlcJhbGciOiIjQzU1NjJ9.eyJhY3NfcGhlbVBiYktleSI6Intclmt0eVwi
OlwiRUNclixclmNydlwiOlwiUC0yNTZclixclnhcljpcikg2bIRnb2g0S1RwR1BxWkZpekxMeEhoNEdCY
kwtUUIZOGY2T3N4U3dRa2dclixclnlcljpcikFIVEpFMzByZFNfZEkyOFJkekFfM2NINVDXTzhFcFpPb3
lOYm0xdXYzc3Ncln0iLCJzZGtFcGhlbVBiYktleSI6IntclmNydlwiOlwiUC0yNTZclixclmt0eVwiOlwiRU
NclixclnhcljpcinNVbIFMZTJXdUEwSlDc01Wek9KX0hNZmplVUVHMDg3WV9ueDIJOUVTTmtclixcl
nlcljpcilVQTGQyNEszNUixUnRKdllreINIZExFS1dQTnZKVXhMb3NIYkZlbfH5dDhcln0iLCJhY3NVUk
wiOiJodHRwczovL3FjY2VydC5ucGNpLm9yZy5pbj9iZXBNL3NhbmRib3hkc3NpbS9hY3NDUmVxQX
BwQXV0aGVudGljYXRpb24ifQ.bn8Jq4q7stUT0vtyCUsYAB_10Q7pzuKln0zjs0Spy9lwOBXTzh7xW
APaiunZWvb6vmpmvFX8twGZYNKUXsZirPvFqMklc3xIGW3tjlZBLD3EystRm9Fku8q_xFdlBOKUm
TnJJwdk5GtHK-eFZWbWBQXdqFSqvheLV5aMxCiDPN8zsls07Y6SNBhcXEG-
cRDxoZMQsZn2i263IgN1okKa8-
3US10LqqZ1unrqsc4l67lQRGa2UzxMWO_LjzCn0w_JVNtGE9KpE-
XVucoZJVOsDWhAMa5JO3nEWtpE0cxCllynP9jugZ7e_qK5FaA5fYAp6mPOLboYO531Hsb8Zfqr3
w",

"authenticationType": "01",

```

"authenticationValue": "AAIRB3InkBdOMhAIGWeQAAACMzQ=",
"dsReferenceNumber": "3DS_LOA_DIS_FTPT_FIME14_1101",
"dsTransID": "03fdd67f-677f-44e0-9357-8ceb5e5e9713",
"eci": "05",
"messageType": "ARes",
"messageVersion": "2.2.0",
"sdkTransID": "c8dfcdf2-2c6a-415d-b520-4506f7f3b9eb",
"threeDSServerTransID": "fa76b3c7-e3e9-54eb-8000-0000020fd9cf",
"transStatus": "C"
}

```

13.1.9RREQ Message from DS – 3DSS (App Flow)

```

{
  "acsTransID": "823012ac-f144-4301-b5f7-f81ef4d412a5",
  "authenticationMethod": "02",
  "authenticationType": "02",
  "dsTransID": "03fdd67f-677f-44e0-9357-8ceb5e5e9713",
  "interactionCounter": "01",
  "messageCategory": "01",
  "messageType": "RReq",
  "messageVersion": "2.2.0",
  "threeDSServerTransID": "fa76b3c7-e3e9-54eb-8000-0000020fd9cf",
  "acsRenderingType": {
    "acsInterface": "02",
    "acsUiTemplate": "01"
  },
  "sdkTransID": "c8dfcdf2-2c6a-415d-b520-4506f7f3b9eb",
  "authenticationValue": "QXV0aGVudGljYXRpb24gdmFsdWU=",
  "transStatus": "Y"
}

```

13.1.10 RRES Message from 3DSS – DS (App Flow)

```
{
  "acsTransID": "823012ac-f144-4301-b5f7-f81ef4d412a5",
  "dsTransID": "03fdd67f-677f-44e0-9357-8ceb5e5e9713",
  "messageType": "RRes",
  "messageVersion": "2.2.0",
  "resultsStatus": "01",
  "sdkTransID": "c8dfcdf2-2c6a-415d-b520-4506f7f3b9eb",
  "threeDSServerTransID": "fa76b3c7-e3e9-54eb-8000-0000020fd9cf",
  "threeDSServerOperatorID": "DFSOperator"
}
```

13.2 Message for ACS

13.2.1 AREQ Message from DS – ACS (APP Flow)

```
{
  "acquirerBIN": "000361603",
  "acquirerMerchantID": "1000",
  "cardExpiryDate": "2608",
  "deviceChannel": "01",
  "deviceInfo":
    "eyJERCI6eyJDMdAxIjoiQW5kcm9pZCIsIkMwMDIiOiJzYW1zdW5nfHxTTS1ONzcxRiIsIkMwMDMiOiJBbmRyb2lkIFRlJUKFNSVNVIDEzIEFQSSAzMyIsIkMwMDQiOiIxMyIsIkMwMDUiOiJlbi1QSCIsIkMwMDYiOiIzMdAiLCJDMdA4IjoiMTA4MHgyNDdAwliwiQzAxMCI6IjE5Mi4xNjguMi4yMyIsIkMwMTEiOiIOMy43NCIsIkMwMTIiOiI0NTkuNTkiLCJDMdEzIjoiY29tLm1vZGlzdW0udGhyZWVkc2ludGVzdGVyIiwiaQzAxNCi6IjJkNWExOTc0LWY4N2MtNGFkNC04MwY4LWViZDU1ODc3MjEzMyIsIkMwMTUiOiI0LjIuMCI6IjIyYiLCJDMdE2IjoiM0RTU0RlNzQzMzI4MjNGRiIsIkMwMTciOiIyMDIOMdEYnTE0NTAyMSIsIkEwMDMiOiIwMSIsIkEwMDYiOiI0JodHRwOi8vd2FwLnNhbnN1bmdtb2JpbGUuY29tL3VhcHJvZi9TQU1TVU5HVUFQUk9GLnhtbCIsIkEwMDciOiI1TVU5HLUFORFJPSUQtTU1TL1NNLU43NzBGliwiQTAwOCi6ImNhliwiQTAwOSI6IjMwMjlyMCI6IjE5MjEzMyIsIkEwMTEiOiIwliwiQ
```

[illegible]

[illegible]

```
"deviceRenderOptions": {
```

```
"sdkInterface": "03",
```

```
"sdkUiType": [
```

"01",

"02",

"03",

"04",

"05"

1

```

},
"dsReferenceNumber": "3DS_LOA_DIS_FTPT_FIME14_1101",
"dsTransID": "03fdd67f-677f-44e0-9357-8ceb5e5e9713",
"dsURL": "http://172.18.248.130:9095/ds/rreq",
"mcc": "5942",
"merchantCountryCode": "246",
"merchantName": "CoffeeHouse - Discover",
"messageCategory": "01",
"messageType": "AReq",
"messageVersion": "2.2.0",
"purchaseAmount": "13100",
"purchaseCurrency": "840",
"purchaseDate": "20240126020726",
"purchaseExponent": "2",
"sdkAppID": "2d5a1974-f87c-4ad4-81f8-ebd558772133",
"sdkEphemPubKey": {
  "crv": "P-256",
  "kty": "EC",
  "x": "sUnQLe2WuA0JWCsMVzOJ_HMfjeUEG087Y_nx9I9ESNk",
  "y": "UPLd24K35B1RtJvYkzSedLEKWPNvJUxLosebFelqyt8"
},
"sdkMaxTimeout": "05",
"sdkReferenceNumber": "3DSSDK74332823FF",
"sdkTransID": "c8dfcdf2-2c6a-415d-b520-4506f7f3b9eb",
"threeDSRequestorAuthenticationInd": "01",
"threeDSRequestorChallengeInd": "03",
"threeDSRequestorID": "1000",
"threeDSRequestorName": "CoffeeHouse - Discover",
"threeDSRequestorURL": "https://www.modirum.com",
"threeDSServerOperatorID": "DFSOperator",
"threeDSServerRefNumber": "3DS_LOA_SER_MOMD_020301_00816",
"threeDSServerTransID": "fa76b3c7-e3e9-54eb-8000-0000020fd9cf",
"threeDSServerURL": "https://discover-dshub.test.modirum.com:9807/dshub/DServer",

```

```
"acctNumber": "****"
}
```

13.2.2 ARES Message from ACS - DS (APP Flow)

```
{
  "acsTransID": "823012ac-f144-4301-b5f7-f81ef4d412a5",
  "acsChallengeMandated": "Y",
  "acsOperatorID": "acsOperator",
  "acsReferenceNumber": "3DS_LOA_ACS_FTPT_FIME14_0009",
  "acsRenderingType": {
    "acsInterface": "02",
    "acsUiTemplate": "01"
  },
  "acsSignedContent":
    "eyJ4NWMIOWsiTUIjRGd6Q0NBbXVnQXkJQkFnSUUpBSUFGZ1ZUK2IWMmtNQTBHQ1NxrR1NJYjNEU
    UVCQ3dVQU1GZ3hDekFKQmdOVkJBWVRBa2xPTVFzd0NRWURWUUVFJREFKVVVRqRVFNQTRHQT
    FVRUJ3d0hRMGhGVGs1QINURU5NQXNHQTFVRUNnd0VUbeJEU1RFTE1Ba0dBmVVFQ3d3Q1N
    WUXhEakFNQmdOVkJBtU1CVVJUSUVOQk1CNFhEVEl6TURreE1UQTJNRGswTVZvWERUSTBNRG
    t4TURBMk1EazBNVm93VORFTE1Ba0dBmVVFQmhNQ1NVNHhDekFKQmdOVkJBZ01BbFJPTVJBd
    ORnWURWUUVFIREfKRFNFV9Ua0ZKTVEwd0N3WURWUUVFLREFST1VFTkpNUXN3Q1FZRFZRUU
    xEQUpKVkRFT01Bd0dBmVVFQXd3RIJGTWdRMEV3Z2dFaU1BMEdDU3FHU0liMORRRUJBVVVBQ
    TRJQkR3QXdnZ0VLQW9JQkFRQ2s1M2tFYVQ2U29SbmdNbFkxdUNrMDI0dFVKR053QUpyakhCO
    HIYVXIPaDhwRklLWDYxNjJ0cUVMTEtpRThJTtNjeG9qZ2lIdHc1d0hFQjAzSWJHaEJBWldXdmFwbl
    pvQzExSHBveVJOV0Rxb0U3ZGpaSkRzU1N5MXlnWW9KY0U1V3Z4a0h4TFpRS05LSXk2RTk1MjQ
    1RIJrbDJuTTJPOVhLOWxOV0E4dUJ6NGpZSW1FNIQwRVRzRDlpanBEa1ZiZH4cXBmd3FoaEVJSkd
    rVUdadml3dIFEclhJVzcybTZkTjhHZURRMEVZQVZ0OU5ES2t4bzZiNzBBY1BZbUpRV2I1N3JXalUz
    QWFTN3hpOEhxTWVxMDExczZ6SFIXamhXclgvRE1EZ1VucC9VL1JxbGpLZENJK3JJeW9sM1BOYW
    xkQU8yb2dPR2ZkbEM5Qzh0Mjd5RmlSQWdNQkFBR2pVREJPTUlwR0ExVWREZ1FXQkJudTh1R
    WNxNndPR0tLRFVXQzN0dmNPUkNDdC96QWZCZ05WSFNNUdEQVdnQIR1OHVfY3E2d09HS0t
    EVVdDM3R2Y09SQ0NOL3pBTUJnTIZiUk1FQIRBREFRSC9NQTBHQ1NxrR1NJYjNEUUVVCQ3dVQUEO
```

SUJBuUFsWHd2bktVUnNMUGJIUktjWjBKRmY3c3N5cHJDcVlzdS84dmkzY0d4dzZpU0QwUjdVdTA
 0VktHXYwTU85a0hzd3pnU2JkOTBkaXRYNWJRReXpqMjVWeFZydnQ0dk1yR0tJUeR3RGNWZDdv
 Q0tNNDI2bDY2L2FHSHRUVGQ2QmFReU4venpUbzNPcWY3RmVTdmNDSkdVcUJlB0l1TGtRQjV6
 MUhZeG5jU2VpS25lOVBoRU9MU0Y3OWFOS091ZzYwd2NmM3pKbVdleG90eVBMWGVcWHVzb
 y94ZEMrNGNUYWZtQ0JRTIFVRUkxU0cwR1czOFpCU2U0ZHRlK1RaSzZGMnBiRW5CZDZ3SE5pSE
 dLbXILQ2txT3FyQ01yT3p5R1hiUFB0THZ3OGtxOXJSdXQvM25rMUICNkp2dUFleFFERTZtM2hyU
 WxlyjNicDY0eVRJUIFDUGFQVYjdlCJhbGciOiJQUz11NiJ9.eyJhY3NfcGhlbVb1YktleSI6Intclmt0eVwi
 OlwiRUNclixclmNydIwiOlwiUC0yNTZclixclnhcljpcikg2blRnb2gOS1RwR1BxWkZpekxMeEhoNEdCY
 kwtUUIZOGY2T3N4U3dRa2dclixclnhcljpcikFIVEpFMzByZFNFZEkyOFJkekFfM2NINvdXTzhFcFpPb3
 IOYmOxdXYzc3Ncln0iLCJzZGtFcGhlbVb1YktleSI6IntclmNydIwiOlwiUC0yNTZclixclmt0eVwiOlwiRU
 NclixclnhcljpcinNVbIFMZTjXduEwSldDc01Wek9KX0hNZmpIVUVHMDg3WV9ueDIJOuVTTmtclixcl
 nlcljpcilVQTGQyNEszNUlxUnRKdlIreINIZExFS1dQTnZKVXhMb3NIYkZlbfHF5dDhcln0iLCJhY3NVUk
 wiOiJodHRwczovL3FjY2VydC5ucGNpLm9yZy5pbj9iZXBNL3NhbmRib3hkc3NpbS9hY3NDUmvXQX
 BwQXV0aGVudGljYXRpb24ifQ.bn8Jq4q7stUT0vtyCUsYAB_10Q7pzuKln0zjs0Spy9lwOBXTzH7xW
 APaiunZWvb6vmpmvFX8twGZYNKUXsZirPvFqMklc3xIGW3tjlZBLD3EystRm9Fku8q_xFdlBOKUm
 TnJlJwdk5GtHK-eFZWbWbQXdqFSqvheLV5aMxCiDPN8zsls07Y6SNBhcXEG-
 cRDxoZMQsZn2i263lgN1okKa8-
 3US10LqqZ1unrqsc4l67lQRGa2UzxMwo_LjzCn0w_JVntGE9KpE-
 XVucoZJVOsDWhAMa5JO3nEWtpE0cxCllynP9jugZ7e_qK5FaA5fYAp6mPOLboYO531Hsb8Zfqr3
 w",

```
"authenticationType": "01",
"dsReferenceNumber": "3DS_LOA_DIS_FTPT_FIME14_1101",
"dsTransID": "03fdd67f-677f-44e0-9357-8ceb5e5e9713",
"eci": "05",
"messageType": "ARes",
"messageVersion": "2.2.0",
"sdkTransID": "c8dfcdf2-2c6a-415d-b520-4506f7f3b9eb",
"threeDSSTransID": "fa76b3c7-e3e9-54eb-8000-0000020fd9cf",
"transStatus": "C"
```

```
}
```

13.2.3 RREQ Message from ACS – DS (APP Flow)

```
{
```

```

"acsTransID": "823012ac-f144-4301-b5f7-f81ef4d412a5",
"authenticationMethod": "02",
"authenticationType": "02",
"dsTransID": "03fdd67f-677f-44e0-9357-8ceb5e5e9713",
"interactionCounter": "01",
"messageCategory": "01",
"messageType": "RReq",
"messageVersion": "2.2.0",
"threeDSServerTransID": "fa76b3c7-e3e9-54eb-8000-0000020fd9cf",
"acsRenderingType": {
  "acsInterface": "02",
  "acsUiTemplate": "01"
},
"sdkTransID": "c8dfcdf2-2c6a-415d-b520-4506f7f3b9eb",
"authenticationValue": "QXV0aGVudGljYXRpb24gdmFsdWU=",
"transStatus": "Y"
}

```

13.2.4 RRES Message from DS -ACS (App Flow)

```

{
  "acsTransID": "823012ac-f144-4301-b5f7-f81ef4d412a5",
  "dsTransID": "03fdd67f-677f-44e0-9357-8ceb5e5e9713",
  "messageType": "RRes",
  "messageVersion": "2.2.0",
  "resultsStatus": "01",
  "sdkTransID": "c8dfcdf2-2c6a-415d-b520-4506f7f3b9eb",
  "threeDSServerTransID": "fa76b3c7-e3e9-54eb-8000-0000020fd9cf",
  "threeDSServerOperatorID": "DFSOperator"
}

```

13.2.5 AREQ Message from DS – ACS (Browser Flow)

```

{

```

```

"acquirerBIN": "000361603",
"acquirerMerchantID": "1000",
"addrMatch": "Y",
"billAddrCity": "Oklahoma City",
"billAddrCountry": "840",
"billAddrLine1": "54 E Pear St",
"billAddrPostCode": "73102",
"billAddrState": "OK",
"browserAcceptHeader":
"text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;
q=0.8,application/signed-exchange;v=b3;q=0.7",
"browserColorDepth": "24",
"browserIP": "174.91.51.140",
"browserJavaEnabled": false,
"browserLanguage": "en-PH",
"browserScreenHeight": "960",
"browserScreenWidth": "1536",
"browserTZ": "300",
"browserUserAgent": "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/537.36
(KHTML, like Gecko) Chrome/120.0.0.0 Safari/537.36",
"cardExpiryDate": "2608",
"cardholderName": "John Smith",
"deviceChannel": "02",
"dsReferenceNumber": "3DS_LOA_DIS_FTPT_FIME14_1101",
"dsTransID": "4b6a43f9-61c4-4e42-a230-d80b5c829776",
"dsURL": "http://172.18.248.130:9095/ds/rreq",
"mcc": "5942",
"merchantCountryCode": "246",
"merchantName": "CoffeeHouse - Discover-3",
"messageCategory": "01",
"messageType": "AReq",
"messageVersion": "2.2.0",

```

```

    "notificationURL": "https://3ds2-mpi-
discover.test.modirum.com/mdpaympi/MerchantServer/msgid/34610411",
    "purchaseAmount": "10000",
    "purchaseCurrency": "840",
    "purchaseDate": "20240131110445",
    "purchaseExponent": "2",
    "shipAddrCity": "Oklahoma City",
    "shipAddrCountry": "840",
    "shipAddrLine1": "54 E Pear St",
    "shipAddrPostCode": "73102",
    "shipAddrState": "OK",
    "threeDSCompInd": "N",
    "threeDSRequestorAuthenticationInd": "01",
    "threeDSRequestorID": "1000",
    "threeDSRequestorName": "CoffeeHouse - Discover-3",
    "threeDSRequestorURL": "https://www.modirum.com",
    "threeDSServerOperatorID": "DFSOperator",
    "threeDSServerRefNumber": "3DS_LOA_SER_MOMD_020301_00816",
    "threeDSServerTransID": "9c220122-fb10-50cb-8000-000002101ceb",
    "threeDSServerURL": "https://discover-dshub.test.modirum.com:9807/dshub/DServer",
    "browserJavascriptEnabled": true,
    "acctNumber": "****"
}

```

13.2.6 ARES Message from ACS – DS (Browser Flow)

```

{
    "acsTransID": "efa74225-be59-4a91-8553-08aa275774b8",
    "acsChallengeMandated": "Y",
    "acsOperatorID": "acsOperator",
    "acsReferenceNumber": "3DS_LOA_ACS_FTPT_FIME14_0009",
    "authenticationType": "01",
    "dsReferenceNumber": "3DS_LOA_DIS_FTPT_FIME14_1101",
    "dsTransID": "4b6a43f9-61c4-4e42-a230-d80b5c829776",

```

```

"eci": "05",
"messageType": "ARes",
"messageVersion": "2.2.0",
"threeDSServerTransID": "9c220122-fb10-50cb-8000-000002101ceb",
"transStatus": "C",
"acsURL": "https://qccert.jaywands.org.in/ds/sandboxdssim/acsCReqBrowserAuthentication"
}

```

13.2.7 RREQ Message from ACS – DS (Browser Flow)

```

{
  "acsTransID": "efa74225-be59-4a91-8553-08aa275774b8",
  "authenticationMethod": "02",
  "authenticationType": "02",
  "dsTransID": "4b6a43f9-61c4-4e42-a230-d80b5c829776",
  "interactionCounter": "01",
  "messageCategory": "01",
  "messageType": "RReq",
  "messageVersion": "2.2.0",
  "threeDSServerTransID": "9c220122-fb10-50cb-8000-000002101ceb",
  "authenticationValue": "QXV0aGVudGljYXRpb24gdmFsdWU=",
  "transStatus": "Y"
}

```

13.2.8 RRES Message from DS – ACS (Browser Flow)

```

{
  "acsTransID": "efa74225-be59-4a91-8553-08aa275774b8",
  "dsTransID": "4b6a43f9-61c4-4e42-a230-d80b5c829776",
  "messageType": "RRes",
  "messageVersion": "2.2.0",
  "resultsStatus": "01",
  "threeDSServerTransID": "9c220122-fb10-50cb-8000-000002101ceb",
}

```

```
"threeDSServerOperatorID": "DFSOperator"
}
```

14 Annexure A – EMVCo Reference Guide

EMV Co Specification	 EMV_3DS_specs_V22 0_122018.pdf
----------------------	--

15 Annexure B – Token Message Extension

For transactions initiated with Token PAN by 3DS Server, 3DS Server is required to add below mentioned message extension.

TAVV – Token Authentication Verification Value

This value is generated by Network Tokenization System (NTS) in response to Token Generation Request from Token Requestor and same (TAVV value) is required by 3DS Server to pass in the AREQ message to DS server for validation.

```
"messageExtension": [
  {
    "name": "tokenization",
    "id": "ID1",
    "criticalityIndicator": true,
    "data": {
      "tavv": "AAGBAFg4ZQAgIxEpESgHAAAAAAI="
    }
  }
]
```

```

    }
  }
]

```

Sample AREQ Message with Token PAN & TAVV

```

{
  "acquirerBIN": "000361603",
  "acquirerMerchantID": "1000",
  "addrMatch": "Y",
  "billAddrCity": "Oklahoma City",
  "billAddrCountry": "840",
  "billAddrLine1": "54 E Pear St",
  "billAddrPostCode": "73102",
  "billAddrState": "OK",
  "browserAcceptHeader":
"text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*
/*;q=0.8,application/signed-exchange;v=b3;q=0.7",
  "browserIP": "174.91.51.140",
  "browserJavaEnabled": false,
  "browserJavaScriptEnabled": true,
  "browserLanguage": "en-PH",
  "browserColorDepth": "24",
  "browserScreenHeight": "960",
  "browserScreenWidth": "1536",
  "browserTZ": "300",
  "browserUserAgent": "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7)
AppleWebKit/537.36 (KHTML, like Gecko) Chrome/120.0.0.0 Safari/537.36",
  "cardExpiryDate": "<TOKEN EXPIRY>",
  "cardholderName": "John Smith",
  "deviceChannel": "02",
  "mcc": "5942",
  "merchantCountryCode": "246",
  "merchantName": "CoffeeHouse - Discover-3",
  "messageCategory": "01",

```

```

"messageType": "AReq",
"messageVersion": "2.2.0",
"notificationURL": "https://3ds2-mpi-
discover.test.modirum.com/mdpaympi/MerchantServer/msgid/34610411",
"purchaseAmount": "10000",
"purchaseCurrency": "840",
"purchaseExponent": "2",
"purchaseDate": "20240131110445",
"shipAddrCity": "Oklahoma City",
"shipAddrCountry": "840",
"shipAddrLine1": "54 E Pear St",
"shipAddrPostCode": "73102",
"shipAddrState": "OK",
"threeDSComplnd": "N",
"threeDSRequestorAuthenticationInd": "01",
"threeDSRequestorID": "1000",
"threeDSRequestorName": "CoffeeHouse - Discover-3",
"threeDSRequestorURL": "https://www.modirum.com",
"threeDSServerRefNumber": "3DS_LOA_SER_MOMD_020301_00816",
"threeDSServerOperatorID": "DFSOperator",
"threeDSServerTransID": "9c220122-fb10-50cb-8000-000002101ceb",
"threeDSServerURL": "https://discover-dshub.test.modirum.com:9807/dshub/DServer",
"acctNumber": "<TOKEN PAN>",
"messageExtension": [
  {
    "name": "tokenization",
    "id": "ID1",
    "criticalityIndicator": true,
    "data": {
      "tavv": "AAgBAFg4ZQAgIxEpESgHAAAAAAI="
    }
  }
]

```

}